

The Welfare Impacts of Large-Scale Land Transactions: Evidence from Ethiopia

Yeshwas Admasu

FAO

Dagim Belay

University of Copenhagen

Workineh A. Kassie

University of Gondar

Jedediah Silver

Hebrew University of Jerusalem

Melkamu W. Tadesse

University of Gondar

Solomon Z. Walelign

Policy Studies Institute

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Abstract

Large-scale land transactions (LSLTs) have expanded rapidly across developing countries over the past decades, especially in Ethiopia. While LSLTs are often promoted as instruments for rural transformation, promising employment creation, technology transfer, and improved market integration, they also raise concerns about displacement, reduced access to natural resources, and weakened customary tenure. We combine two analyses to study the medium-run effects of LSLTs in Ethiopia. First, we leverage the plausibly exogenous timing of LSLT entry to estimate the effects on local economic and environmental outcomes. Second, we conduct an original survey of households in Gambela region—an LSLT hotbed—including detailed questions on households’ interactions with LSLTs to shed light on mechanisms. Here we exploit the fact that potential investors relied on observable factors such as soil characteristics, road access, and proximity to conflict zones to identify causal effects using double-debiased machine learning. For a subsample of Gambela, we also observe whether the LSLT simply received a concession or actually engaged in cultivation, which allows us to distinguish the effects of an active investment from those of cordoning off a large tract of land. At the national level, we find modest effects on crop productivity along with accelerated deforestation, albeit with no effect on overall tree cover and significant reductions in burned area. In Gambela, we find similar effects on crop burning but also that LSLTs reduced agricultural productivity and tree cover. We estimate a 1.2 percentage point increase in the probability of engaging in wage employment for every 1,000 ha of LSLT within 5–10 km—driven solely by active LSLTs—but no effects on other labor market outcomes, agricultural production or consumption. The areas surrounding cultivated parcels also experience reduced deforestation, while there are no economic effects surrounding uncultivated parcels. These results are consistent with limited labor market and other direct linkages between LSLTs and households and suggest that both the main promises and pitfalls of LSLTs may be overstated in such contexts.

1 Introduction

Large-scale land transactions (LSLTs) have become increasingly common in developing countries, with millions of hectares transferred from governments to private investors in recent years. These transfers have been promoted as a win-win, allowing uncultivated land to be put to productive uses while providing labor market opportunities, technology transfers and aggregate demand increases to host communities. However, such transactions have often been referred to as “land grabs” due to reports of illegally expropriating farmland, environmental degradation and deforestation (Anti, 2021). Identifying these positive and negative potential effects, which may occur simultaneously, and their incidence is therefore crucial for policy. However there is limited credible causal evidence on these channels: In particular, well-identified analyses with administrative panel data often lack the ability to directly observe how households interact with and are affected by LSLTs.

We study the potential for both of these channels in Ethiopia by combining two research designs: a national-level event-study leveraging plausibly exogenous timing in the entry of LSLTs across woredas (districts) and an ex-post analysis using original survey data of households in Gambela region, which was highly exposed to LSLTs. In particular, our survey asks farmers to detail their direct interactions with LSLTs—including land conflict and expropriation, employment, technology transfer, and other economic transactions. This also allows us to estimate heterogeneous treatment effects with regards to proximity to LSLTs, which is crucial for understanding their distributional effects.

At the national level, evidence of agricultural gains is at best suggestive: net primary productivity (NPP), a remote-sensed measure of vegetation productivity, rises by +0.0062 ($p = 0.101$), with the gain concentrated in woredas receiving three or more transfers, and survey data show no effects on yields or input use apart from a decline in irrigation. Environmental margins move more sharply: forest loss rises by 0.45 percentage points of woreda area per year (+0.0045***, SE 0.0014)—several times the area actually transferred when cumulated, and concentrated in forest-frontier woredas—while burned area falls (-0.0096**), consistent with the displacement of fire-based land management.

In Gambela, our more detailed administrative data and original survey data allow us to examine a wider set of outcomes more granularly. Our main specification examines the effects of LSLT exposure on grid cells cropped at baseline within 5 km of a concession. Here, burned area falls sharply (-0.1113***), tree cover declines (-0.609*** percentage points), and vegetation indices fall modestly (NDVI -0.0072***), with no detectable effect on forest loss. Turning to the household survey, we find very little evidence of economic impacts—only a small increase in the probability of wage employment. To make sense of these results, it is worth keeping in mind that only about 20% of allocated LSLTs in Gambela actually engaged in cultivation, compared to over 80% in Amhara (Eisfelder et al., 2024).¹ If it is plausibly exogenous whether a community was exposed to an implemented vs. non-implemented investment—which parallel trends on time-varying outcomes and conditioning on observables suggest—then we can separate the effects of exposure to an actual investment from the effects of the enclosure of a large parcel of land. Consistent with this, we find that the effects on wage employment are driven by implemented investments. However, even realized

¹It is common knowledge that many investors simply used the land title as collateral for real estate loans outside of Gambela without intending to operationalize an agricultural investment.

investments do not affect overall income or consumption in the long run, suggesting that exposure to LSLTs had little effect on local communities in Ethiopia.

Our study makes two major contributions. First, we contribute to a growing but methodologically limited literature on the economic impacts of LSLTs. A meta-analysis by Akalu et al. (2025) identifies 143 relevant studies, of which only 39 are quantitative; among these, 74% rely on propensity-score matching with cross-sectional data, where matching on post-treatment covariates raises identification concerns. Table 11 summarizes the key quantitative studies. Even the three DiD studies use two-way fixed effects, despite the compounding effects of early-treated units in later periods we find representing the classic case in which negative weighting bias occurs. Our study addresses this gap by exploiting the staggered rollout of Ethiopia’s LSLT program across woredas, using heterogeneity-robust DiD estimators—primarily the Callaway-Sant’anna estimator (Callaway & Sant’Anna, 2021) that accommodate staggered adoption with treatment effect heterogeneity. For the Gambela survey, which is by necessity a post-treatment cross sectional analysis, we improve upon PSM by applying state of the art DML (Chernozhukov et al., 2018). Importantly, we can validate this against remote-sensed outcomes for which we do have baseline data.

Second, our original survey of 2,389 households in Gambela provides a unique look inside the “black box” of how households interact with LSLTs. Directly observing these reported interactions—which include limited labor market engagement and few other economic linkages—provides a lens through which to interpret our modest to moderate treatment effects, which are non-monotonic in distance to LSLTs. In particular, we interpret these as reflecting costs of direct proximity to LSLTs despite positive ambient treatment effects. We believe that we are the first to document this with a rigorous quantitative framework.

2 Background: Ethiopian Land Policy and Large-Scale Agricultural Investment

2.1 Evolution of Large-Scale Agricultural Land Investment Policy

For most Ethiopians, agricultural land is the foundation of livelihoods, contributing significantly to household welfare both directly and indirectly. Agricultural land has also been one of the defining factors of social status and, in some cases, political power throughout Ethiopian history: families with large landholdings have historically been associated with high social and political standing. The role of agricultural land has remained crucial in modern Ethiopia as it is a major source of raw materials and direct input to the growing non-agricultural sectors (manufacturing and services), although the contribution of the agricultural sector to the economy is diminishing. Hence land policies and reforms have taken center stage in Ethiopian agricultural policy, particularly in the modern era. Historically, Ethiopia has formulated a number of land policies and undertaken a number of land reforms with the objectives of bringing about suitable concurrent economic and political transformation (Dunning, 1971), often in response to public demand and changes in government (Kebede, 2008). Hence, land reform has been seen as one of the prerequisites for national growth (Dunning, 1971).

During the imperial regime’s reign to 1974, the Derg military regime from 1974 to 1991,

and the post-Derg regime since 1991, Ethiopia has experienced various land tenure arrangements (Bodurtha, 2011). Historically, large farm investment projects have been in operation since the imperial period in Ethiopia (Ali et al., 2019). However, the approaches, priorities, and intensity of investments vary between regimes. Although cash crop production under private investors was prioritized before the 1974 revolution, the Derg regime nationalized large-scale private commercial farms and converted them into state-produced staple foods until the Ethiopian People’s Revolutionary Democratic Front (EPRDF) took power in 1991 (Ali et al., 2017; Kassie et al., 2024).

In the imperial regime, land regulations allowed for the transfer, market, and mortgage of private properties, allowing for a more equitable allocation of land among landowners in different places (Gebregziabher & Holden, 2011; Beyene Leta et al., 2021). In this regime, the country experienced two land tenure systems: the *rist* and *gult* (Teklu, 2004). The communal *rist* system divided and distributed land by the community according to lineage claims (Bodurtha, 2011). *Gult* is the non-hereditary claim to all, or a portion of the tribute owed to the king by the land’s occupier (Dunning, 1971; Gebregziabher & Holden, 2011; Beyene Leta et al., 2021).

The Derg regime’s Land to the Tiller land reform in February 1975 eliminated all feudal land laws and turned land over to the public (Bodurtha, 2011; Ambaye, 2015). The February 1975 land reform declaration nationalized all rural land, replacing the imperial period’s age-old land tenure with a system in which the state owns all land and gives it to farmers on a use-right basis. State farms were created from commercial large-scale modern farms. On about 800 hectares, peasant associations were formed to allot land to farmers in their authority based on family size (Kebede, 2008). The Derg nationalised rural and urban land in 1975 and redistributed land with a maximum holding of 10 hectares per farm (Holden & Yohannes, 2002). However, the regime’s practices resulted in unintended consequences such as farmland fragmentation through redistribution, eviction of rural populations through forcible villagization, the inefficiency of farmer cooperatives, failure of agricultural productivity, and total poverty (Kebede, 2002; Bodurtha, 2011).

Following the fall of the Derg regime, the newly formed transitional government led by the EPRDF announced the continuation of the Derg regime’s land policy through an amendment, and state ownership of land was established in the new constitution in 1995 (Deininger et al., 2003; Hagmann, 2006). Land lease limits, labor hiring on a private farm, sharecropping, and individuals holding more than 10 ha were all banned under the present land policy. Government allocation, inheritance, gift, and land leasing are all options for obtaining rural land, although it cannot be sold, swapped, or mortgaged (Bodurtha, 2011).

In Ethiopia, large-scale agricultural land investment was officially introduced during the third (1968–73) five-year plan of the imperial regime. Several investors rented land from the government and developed commercial farms until the Derg confiscated commercial farms to manage and expand state farms, which were abandoned meaningfully post-1991 when the EPRDF recognized the decisive role of private capital expansion in the development of large-scale modern farming (Ali et al., 2017; Kassie et al., 2024). Agricultural Development-Led Industrialization (ADLI) designed in 1993 prioritized enhancing the productivity of small-holder farming and expansion of large-scale private commercial farms. Private agricultural investments were intensified during the first 5-year period of the GTP (2010–2015), when the Ethiopian Agricultural Investment Land Administration Agency was established in 2013.

Large-scale land transactions in Ethiopia have been increasing, particularly in the low-land regions of the country. These areas, traditionally inhabited by pastoralist and agro-pastoralist communities, have historically experienced limited state control and governance (Moreda, 2015). Much of the land offered is classified by the state as ‘unused’ or ‘under-utilised’, overlooking the spatially extensive use of land in shifting cultivation and pastoralism (Moreda, 2017). Most of the earmarked and transferred agricultural land occurs in four of the nine regional states: Benishangul-Gumuz, Gambela, Oromia, and Southern Nations Nationalities and Peoples (SNNP), with greater intensity in the Benishangul-Gumuz and Gambela regions. The government suddenly suspended the program in 2016 without transferring all land designated for large-scale agricultural investments (Hajjar et al., 2020; Moreda, 2017).

2.2 Dynamics of Large-Scale Land Investment in Ethiopia

The Ethiopian government designated over 11.5 million hectares of land for large-scale investment purposes. Between 2005 and 2015, approximately 2.47 million hectares were transferred to investors, primarily from the earmarked land. The second Growth and Transformation Plan (GTP II, 2015/16–2019/20) targeted an increase in total area of land verified for large-scale agricultural investment from 1.1 million hectares in 2014/15 to 2.443 million hectares, with nationwide total land transfer projected to reach 4.315 million hectares by the end of 2020 (MoFED, 2015).

In the Gambela region, a peak in transfers of land occurred between 2012 and 2015 (MoFED, 2015), aligning with Ethiopia’s first Growth and Transformation Plan (GTP I). This was followed by a steep decline post-2016 due to the government’s abrupt suspension of the land transfer program (Hajjar et al., 2020; Moreda, 2017; Rulli & D’Odorico, 2014). In Gambela, large-scale land transfers have occurred in eight of the region’s thirteen woredas. The Anuak Zone leads in both the number of investors and total land area leased, followed by the Nuer Zone and the Majang Zone. At the woreda level, Gambela Zuria Woreda holds the highest number of investors (200), followed by Itang Special Woreda (137) and Abobo Woreda (130).

3 Data

Our analysis draws on five data sources, summarized in Table 1. Summary statistics for the main outcomes and treatment measures from each source are provided in Tables 12-14.

3.1 Treatment: LSLT Parcels and Timing

Treatment is defined by the timing of LSLT entries at the woreda (district) level nationally, and at the kebele (ward) or pixel level in Gambela. We use the universe of approved investments from the Ministry of Agriculture, which provides information on the woreda, size, year of entry and name of the investor for each LSLT. This records 147 treated woredas with first treatment between 1995 and 2016. Total LSLT area across all treated woredas is 966,026 hectares, with a median of 650 ha per woreda. In Gambela, we use the shapefile with the exact geocoordinates of LSLT parcels. This covers 36 treated kebeles (treatment

years 2007–2014), which we match to the Ministry of Agriculture records to obtain the year of treatment. Figures 1(a) and 1(b) maps the spatial and temporal distribution of treatment at both levels.

3.1.1 GIZ Data

To distinguish LSLTs that entered production from those that exist only on paper, we use data from the GIZ Remote Sensing for Agricultural Investment Monitoring (RS4AIM) project, which mapped large-scale agricultural investment (LSAI) parcels and their cultivation status in Gambela, Benishangul-Gumuz, and Amhara—roughly 60% of LSAI parcels in Ethiopia by count (see Eisfelder et al., 2024). The data comprise parcel polygons (450 in Gambela, 211 in Benishangul-Gumuz, and 67 in Amhara) together with ground-truth field campaigns conducted in October–November of 2021 and 2022, in which surveyors recorded land cover and crop type at over 5,700 GPS points. We classify a parcel as verified if at least one ground-truth survey point falls within its boundary—in practice a proxy for active cultivation, as 96% of in-parcel observations record growing crops. In Gambela, 99 of 450 parcels are verified, consistent with Eisfelder et al. (2024), who find that only about 20% of parcels in the region were actually cultivated.² Because the field campaigns surveyed accessible parcels rather than a random sample, we examine below how verified and unverified parcels compare on predetermined soil and locational characteristics and pre-treatment trends. The Benishangul-Gumuz parcels additionally carry administrative registration years, which will allow ongoing work to extend the actual-vs-planned analysis beyond Gambela with treatment timing measured independently of the Ministry of Agriculture records.

3.2 Remote-Sensing Outcomes (GEE)

We construct woreda-year, kebele-year, and pixel-year panels of environmental outcomes from Google Earth Engine for 2001–2023. Vegetation productivity is measured by the Normalized Difference Vegetative Index (NDVI), the Enhanced Vegetation Index (EVI), and annual net primary productivity (NPP); tree cover by the MODIS Vegetation Continuous Fields (VCF) percent tree cover; forest loss by the Hansen Global Forest Change annual loss layer at 30m resolution (Hansen et al., 2013); and fire activity by annual burned-area fraction. We also extract annual land-cover class shares from MODIS, restricting to forest, savanna, grassland, and cropland. Pixel-level values are extracted at native sensor resolution (30–500m depending on the product) and aggregated to administrative boundaries as area-weighted means. Cropland NDVI is masked to 2001-baseline cropland pixels to avoid conditioning on endogenous land-use change, and annual precipitation (CHIRPS) serves as a control.

3.3 Agricultural Sample Survey (AgSS)

Agricultural outcomes come from the Central Statistical Agency’s Agricultural Sample Survey (AgSS), a nationally representative repeated cross-section of smallholder field-level data

²Balance and trend tests (Tables 23 and 24) use the 330 parcels large enough to contain at least one remote-sensing extraction pixel with complete 2001–03 baseline data, of which 78 are verified.

collected each Meher season. We use crop-cut yields, cultivated area, and input use—fertilizer, improved seed, and irrigation—along with reported crop damage, and aggregate to woreda-year means for 905 woredas over 2003–2020. Because enumeration areas are re-drawn across rounds, the AgSS woreda panel is unbalanced. Survey years are recorded in the Ethiopian calendar and converted to Gregorian years to align with the treatment and remote-sensing panels.

3.4 Gambela Household Survey

We conducted an original household survey covering 2,389 households across 103 kebeles in Gambela region, with two waves (2022–2023). It includes standard consumption and income modules alongside detailed questions on household interactions with LSLTs: employment, land transactions, equipment rental, and perceptions. Treatment exposure is measured as the total LSLT area and count within distance rings (0–5, 5–10, 10–20 km) around each household.

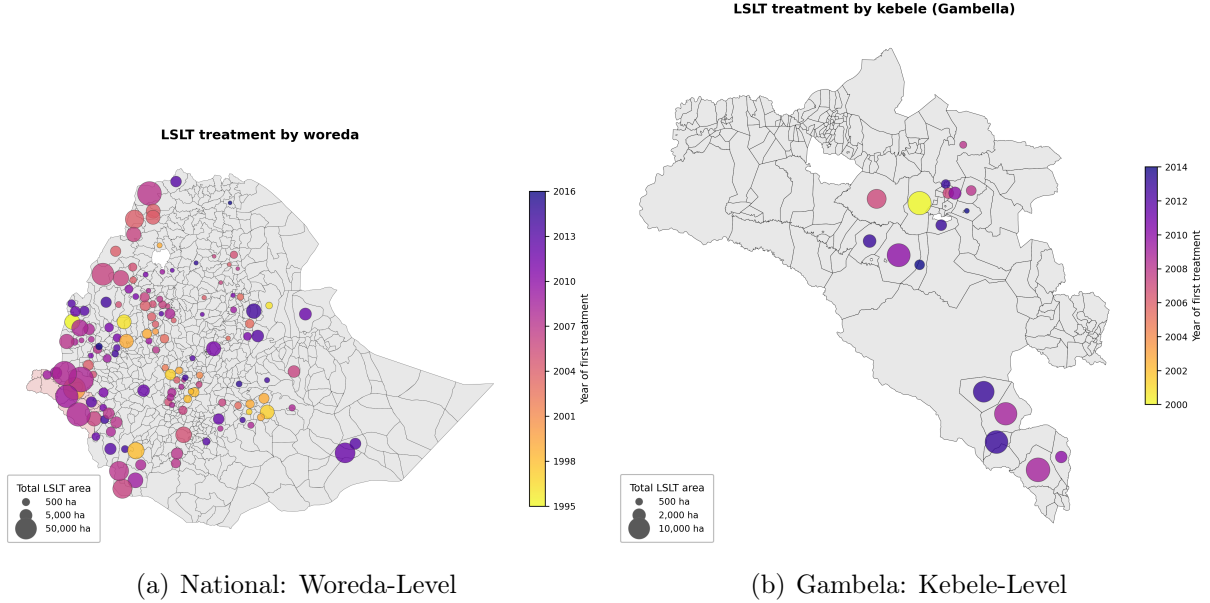


Figure 1: Spatial and temporal distribution of LSLT treatment. Circle color indicates year of first treatment (darker = earlier); circle size is proportional to total LSLT area (hectares). Grey polygons are untreated administrative units.

4 Empirical Strategy

We leverage two main identification strategies to estimate the causal effects of LSLTs on local economies and the environment. First, we exploit the plausibly exogenous timing of LSLT entry at the woreda (district) level across Ethiopia in an event study design. We are also able to conduct the event study at the more granular kebele (ward) level in Gambela Region, where we have the exact geolocations of LSLTs.

Table 1: Data Sources and Outcome Variables

Variable	Source	Resolution	Period	Unit	Level
<i>Panel A: GEE Remote-Sensing Outcomes</i>					
NPP	MODIS MOD17A3HGF	500m	2001–2023	kgC/m ² /yr	Woreda, Kebele
Cropland NDVI	MODIS MOD13A1	500m	2001–2023	Index (0–1)	Woreda
NDVI (raw)	MODIS MOD13A1	500m	2001–2023	Index (0–1)	Woreda, Kebele
Forest Loss	Hansen GFC v1.12	30m	2001–2023	Fraction	Woreda, Kebele
Tree Cover (VCF)	MODIS MOD44B	250m	2000–2020	% cover	Woreda, Kebele
EVI	MODIS MOD13A1	500m	2001–2023	Index (0–1)	Woreda, Kebele
Burned Area	MODIS MCD64A1	500m	2001–2023	Fraction	Woreda, Kebele
<i>Panel B: Land Cover Shares (Gambella Kebele Level)</i>					
Forest Share	MODIS MCD12Q1	500m	2001–2022	Fraction	Kebele
Cropland Share	MODIS MCD12Q1	500m	2001–2022	Fraction	Kebele
Savanna Share	MODIS MCD12Q1	500m	2001–2022	Fraction	Kebele
Grassland Share	MODIS MCD12Q1	500m	2001–2022	Fraction	Kebele
<i>Panel C: Agricultural Sample Survey (AgSS) Outcomes</i>					
Yield	CSA AgSS	Field survey	2003–2020	kg/ha	Woreda
Cultivated Area	CSA AgSS	Field survey	2003–2020	log ha	Woreda
Fertilizer Use	CSA AgSS	Field survey	2003–2020	Share	Woreda
Improved Seed	CSA AgSS	Field survey	2003–2020	Share	Woreda
Irrigation	CSA AgSS	Field survey	2003–2020	Share	Woreda
Crop Damage	CSA AgSS	Field survey	2003–2020	Share	Woreda
<i>Panel D: Gambella Household Survey (LSIT) Outcomes</i>					
Log Consumption	LSIT Survey	Household	2022–2023	Log Birr	HH
Log Wage Income	LSIT Survey	Household	2022–2023	Log Birr	HH
Log Crop Income	LSIT Survey	Household	2022–2023	Log Birr	HH
Log Harvest (kg)	LSIT Survey	Household	2022–2023	Log kg	HH
<i>Panel E: Treatment Data</i>					
LSLT Parcels	MoA/AG_INV	Polygon	1995–2016	Hectares	Parcel
Woreda Boundaries	CSA 2013	Polygon	—	—	Admin
Kebele Boundaries	CSA 2007	Polygon	—	—	Admin
<i>Panel F: GIZ Investment Monitoring (RS4AIM)</i>					
LSAI Parcels	GIZ RS4AIM	Polygon	2021–2022	Hectares	Parcel
Cultivation Ground Truth	GIZ RS4AIM	GPS point	2021–2022	Crop type	Point
Verified Status	GIZ RS4AIM	—	2021–2022	Binary	Parcel
Registration Year (BG)	GIZ RS4AIM	—	2007–2021	Year	Parcel

Notes: GEE outcomes are extracted at native sensor resolution and aggregated to administrative boundaries (woreda or kebele) as area-weighted means. Cropland NDVI is masked to 2001-baseline cropland pixels (IGBP classes 12, 14) to avoid endogenous land-use change. Land cover shares use the IGBP classification from MODIS MCD12Q1. AgSS outcomes are aggregated from field-level survey data to woreda-year means. The LSIT household survey covers Gambella region with two rounds (2022–2023) across 103 kebeles. GIZ RS4AIM data cover large-scale agricultural investment (LSAI) parcels in Gambella (450), Benishangul-Gumuz (211), and Amhara (67); a parcel is verified if the RS4AIM field campaigns (Oct–Nov 2021 and 2022) recorded crop observations within its boundary. Benishangul-Gumuz parcels carry administrative land-registration years (converted from the Ethiopian calendar), providing treatment timing independent of MoA records.

Second, we conduct a cross-sectional analysis using our original survey of 2,389 households in Gambela. Since both waves of these survey are post-treatment, we employ a selection-on-observables design using double-debiased machine learning techniques (Chernozhukov et al., 2018), leveraging the fact the LSLT siting decisions were largely made on observable characteristics such as soil quality and road distance. While this requires stronger identification assumptions, the results are largely consistent with those from the event study analysis, reinforcing the credibility of both sets of findings. Moreover, the survey provides a unique look inside the “black box” of the interactions between households and LSLTs.

4.1 Event Study

Here we leverage the plausibly exogenous timing of LSLT entry into woredas across Ethiopia. While investors have some flexibility in choosing investment locations, the investment is subject to a long series of bureaucratic approvals from the Ethiopia Investment Commission (EIC), the Ministry of Agriculture (MOA) and regional and local authorities. Therefore, conditional on a woreda being targeted for investment, idiosyncratic variation in the time to approval arguably provides suitable identifying variation.

However the staggered, continuous, and repeated nature of our treatment—multiple LSLTs of varying sizes enter woredas over time—presents numerous challenges flagged by the recent literature on negative-weighting bias in two-way fixed effects (TWFE) setups (see Goodman-Bacon, 2021; Roth et al., 2023; Baker et al., 2025). To account for the repeated nature of treatment, we adapt the approach of de Chaisemartin and D’Haultfœuille (2024) and pool all admissible comparisons of woreda-years that experienced an LSLT entry to those with identical treatment histories that did not. This allows us to use the full set of LSLT entries in the data (as opposed to just the first) and also estimate marginal treatment effects at different numbers of LSLTs without contamination from the dynamic effects of earlier treatments. We implement this by stacking two-by-two (Callaway & Sant’Anna, 2021)-style blocks of woreda-years receiving a k -th LSLT entry against those that experienced $k - 1$ previous entries in the same years but did not (yet) receive a k -th entry. Our main specification pools estimates over different values of k , while we also report marginal treatment effects by k . In our main specification, we restrict the sample to ever-treated woredas—exploiting the assumption of random treatment *timing* and dropping never-treated woredas which may be poor comparisons³—showing robustness to including the full sample and using MoA-designated potential investment areas (PIAs) as controls.

The continuous (in terms of investment size) treatment further complicates matters, requiring stronger assumptions on how units with a higher treatment dose would have evolved had they received a lower dose (Callaway et al., 2024). This restricts the set of estimands for which we can cleanly identify a causal effect. In our main specifications, we consider the approval of an LSLT over 20ha as a binary treatment. As a robustness check, we weight treated woredas by the area transferred to LSLTs

Moreover, the fact that many approved investments never materialized means that our estimates should be interpreted as intent-to-treat (ITT) estimates.⁴ However, for parts of

³For example, woredas in treeless desert regions cannot have parallel trends on deforestation.

⁴Ongoing analysis is using deep-learning methods applied to Sentinel-2 satellite imagery to identify the establishment of physical LSLTs in treated pixels (following methods developed by Sy & Gehrke, 2026),

Gambela, Benishangul Gumuz and Amhara, we have ground-truth on whether or not an LSLT parcel was actually being cultivated based on fieldwork by (Eisfelder et al., 2024). We leverage these additional data in an actual-vs-planned design for Gambela.

4.1.1 Gambela Event study

We also leverage our more complete administrative data from Gambela—including the precise geolocations of LSLT parcels—to estimate the effects of a more granularly-defined treatment. Following Kassie et al. (2024), we can also compute pixel-level estimates for remote-sensed outcomes, although we are interested in their effects on households near the concession rather than deforestation on the concession itself. In particular, we extend their analysis by defining a cell as treated if an LSLT has entered within a 5km radius of the cell’s boundary and restrict the sample to cells cropped at baseline (2001). In addition, this analysis helps serve as a validation of the selection on observables approach we employ below—in particular, if parallel trends hold for remote-sensed outcomes, one may find the selection on observables assumptions employed below more plausible.

In Section 7.3 we also present results aggregated to the kebele level. This serves as a different way of aggregating treatment effects across space than the 5km buffer zones, which is more similar to the woreda-level estimates at the national level. There are 198 kebeles and 13 woredas in Gambela. However, this is only feasible for the remote-sensed outcomes, as we do not have sub-woreda identifiers for the AgSS.

4.2 Gambela Survey Analysis

Since LSLTs under the government policy stopped in 2016, seven years before data collection, we employ a selection-on-observables design to estimate the causal effect of LSLTs on local economic outcomes. Selection-on-observables designs need to assume that treatment assignment is as good as random conditional on observable covariates. While these assumptions would be difficult to justify ad hoc, investors reportedly based their choice of land almost entirely on soil tests, distance to roads and security. Fortunately, precise geolocated data exists for these variables, allowing us to observe much of the key information upon which they based their locations.

Of course, we don’t know how firms used this information to make their location decisions. Since this may have involved a complicated function of a high-dimensional vector of covariates, Double-Debiased Machine Learning (Chernozhukov et al., 2018), hereafter DML, is a natural way to flexibly capture these selection patterns. DML applies orthogonalization and sample-splitting to overcome regularization bias and overfitting, which are common issues in machine learning. As is common, we assume a partially linear model of treatment D_i effects on outcome Y_i . D_i may either be a dummy variable for the allocation of land in a kebele to an LSLT or a continuous measure, such as the share of land in a kebele allocated. We allow outcomes and treatment assignment to depend on (pre-determined) covariates X_i through deterministic nuisance functions g_0 and m_0 respectively.

which would provide valid average treatment on the treated (ATT) estimates if an exogenous component of compliance can be isolated, which we believe to be plausible in this setting.

$$\begin{aligned} Y_i &= \beta D_i + g_0(X_i) + \varepsilon_i \\ D_i &= m_0(X_i) + \nu_i \end{aligned} \tag{1}$$

Under the assumptions that $E[\varepsilon_i|D_i, X_i] = 0$ and $E[\nu_i|X_i] = 0$, the DML procedure provides consistent estimates of the treatment effect, β , using standard machine learning methods to estimate g_0 and m_0 . In our case, m_0 is the true relation firms use to decide whether to invest in a site as a function of geophysical and other location-specific factors and g_0 is these factors' direct effects on outcome Y_i . For this procedure to be valid, we assume that we observe the attributes that firms used to choose where to invest and that our estimate $\hat{m}$ well approximates the decision process.

Threats to identification would include missing relevant geophysical attributes, or if LSLTs selected locations based on the receptiveness of local communities or the availability of local labor. We believe that this is unlikely to be the case. First, we have comprehensive soil maps of the area, both from the Ethiopian government and high-resolution gridded models. Second, both government officials and communities were not extensively consulted about LSLTs and given the low engagement of LSLTs with local labor markets, it's unlikely that labor availability played much of a role in siting decisions.

4.3 Actual vs. Planned: Gambela

Given that only about 20% of parcels in a sub-sample of Gambela analyzed by (Eisfelder et al., 2024) were found to actually be cultivated, the results from the specification above could be interpreted as a diluted ITT. If whether a parcel is cultivated is orthogonal to potential outcomes of nearby pixels/households, one can test whether the effects in areas surrounding cultivated parcels differ from those surrounding uncultivated parcels. Importantly, one might expect economic spillovers to be most relevant for LSLTs that actually engage in production. However, it is worth noting that an uncultivated LSLT is not a pure placebo treatment. The land is still transferred to the investor, who may enclose it, prevent the local population from access it, and develop it for other purposes than crop cultivation. In contrast, one can interpret a cultivated LSLT as a bundle of the transfer/enclosure treatment and real agricultural production.

We show in section 5 that cultivated and uncultivated parcels exhibit parallel trends on time-varying remote-sensed outcomes. This supports the notion that whether a household was exposed to a materialized LSLT or simply a demarcated concession is as good as random by reducing concerns that investors' decisions to cultivate a parcel was influenced by time-varying factors related to nearby communities. For survey outcomes, which do not have baseline data, we must extend our selection on observables assumption—that investors chose land to acquire based on soil characteristics, location and proximity to conflict—to the decision to cultivate it. With these identifying assumptions, we can test whether the effects we observe can be attributed to the establishment of an active LSLT or simply the transfer of the land. We therefore run the analyses in subsection 4.1.1 and subsection 4.2 on the subsample of parcels in Gambela classified by GIZ, redefining D_i based on exposure to cultivated parcels.

5 Results

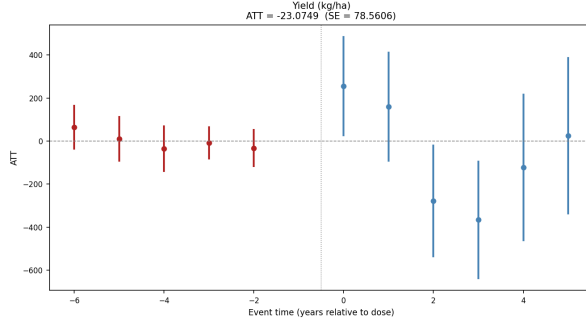
5.1 National Level Event-Study

We estimate the effect of large-scale land transfers on remote-sensing and agricultural outcomes at the woreda level using a staggered difference-in-differences design that pools all admissible treatment doses (see Section 4). Our main specification uses the sample of ever-treated woredas, while we also show robustness to using the full sample of woredas and woredas designated as “potential investment Areas” by the Ministry of Agriculture but not treated as controls. Woredas with any LSLT parcel pre-2001 are dropped, as they are never clean controls. We use design-adjusted standard errors (Abadie et al., 2020), which account for the fact that we observe a non-trivial share of woredas in Ethiopia (see the notes to Table 3).

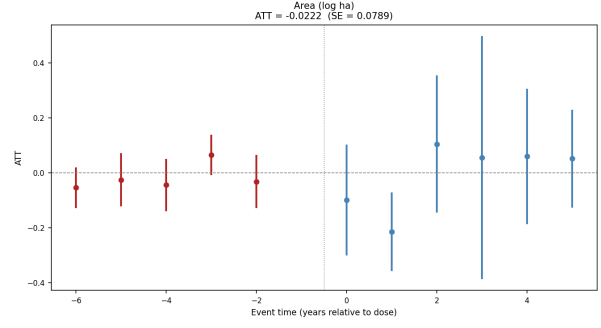
In Figure 2 and Table 2, we look at effects on agricultural outcomes from the AgSS. The only robust effect we see is a decrease in the share of farm households using irrigation. By the fifth year after treatment, the effect is on par with the average pre-treatment irrigation rate of about 2%. This is consistent with LSLTs competing with local farms for irrigation water access. Within five years of entry, we see a significant negative effect on the share of households using improved seed and crop damage, while yields first increase and then decrease relative to not yet treated woredas before the effect disappears. However, in each of these cases, the pooled coefficient over five post-periods is insignificant. We also do not observe significant effects on area cultivated or fertilizer use.

We then turn to remote-sensed outcomes, which relate to both agricultural productivity and land use. In terms of productivity, we see an eventual small increase in net primary productivity (NPP), whose effects mirror the shape of the effect on yields in the AgSS. However, we do not see any effect on crop NDVI or EVI overall. In terms of environmental outcomes, we see a significant increase in forest loss, which cumulates to roughly 20% of the woreda’s forest stock and is several times larger than the median LSLT concession. The effect is also nearly double in woredas with dense forest at baseline. However, these effects are not large enough to affect overall tree cover, as measured by VCF. While this could be attributed to differences in measurement, it suggests that much of the affected forest area is being replaced. Moreover, burned area falls by -0.0096^{**} (SE 0.0039), consistent with fire-based land management being displaced where investments arrive.

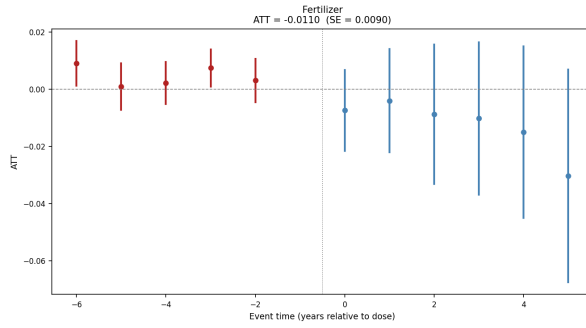
These pooled effects mask some heterogeneity in the dose response function, as we are able to estimate marginal treatment effects of each additional LSLT in Table 2 and Table 3. We find that the effects on forest loss are entirely driven by the first LSLT entry, while VCF falls for 3rd and subsequent doses. In addition, productivity measures (NPP, NDVI, EVI, and evapotranspiration) increase from the third dose. In the AgSS, we see no significant effects from first doses, but a decrease in yields and an increase in area from the 3rd dose onwards. These patterns from later doses could reflect effects that ratchet with treatment intensity, or heterogeneity in the woredas with a propensity to receive repeated investments, which by construction occur later in the sample. Notably, 3rd-dose effects are estimated from 36 woreda-year events for remote-sensed outcomes and 19 events in the AgSS sample. In either case, it becomes clear that a single LSLT entry has little effect on environmental and agricultural outcomes at the woreda level beyond deforestation.



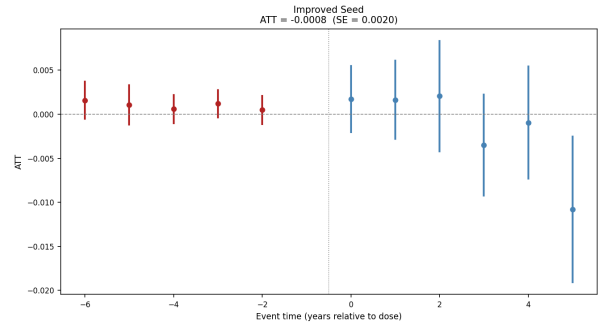
(a) Yield (kg/ha)



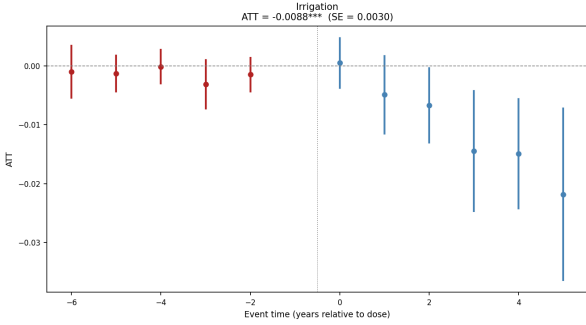
(b) Area (log ha)



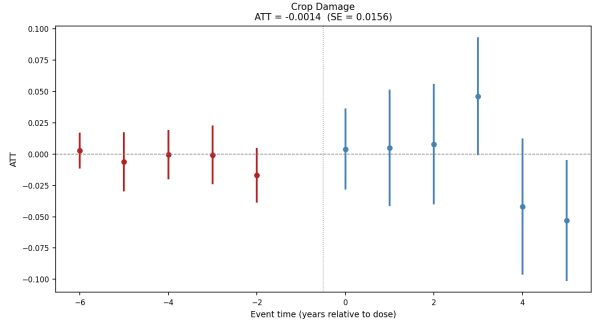
(c) Fertilizer



(d) Improved Seed



(e) Irrigation



(f) Crop Damage

Figure 2: Event-study estimates pooling all admissible LSLT doses: AgSS agricultural outcomes. Design as in Figure 3; the AgSS panel is an unbalanced repeated cross-section, which the block estimator accommodates by requiring each woreda only at a block's two endpoint years. See Table 2 for effects by dose number.

5.1.1 Robustness

The appendix (Section 7.5) additionally reports the corresponding first-treatment Callaway–Sant’Anna battery, which the pooled estimator nests as its first-dose component: the main first-entry specification (Tables 15 and 16), alternative control pools including the full sample with never-treated controls and woredas designated as potential investment areas but never treated, and a second-entry specification following de Chaisemartin and D’Haultfoeuille (2024) (Figure 9 and table 17).

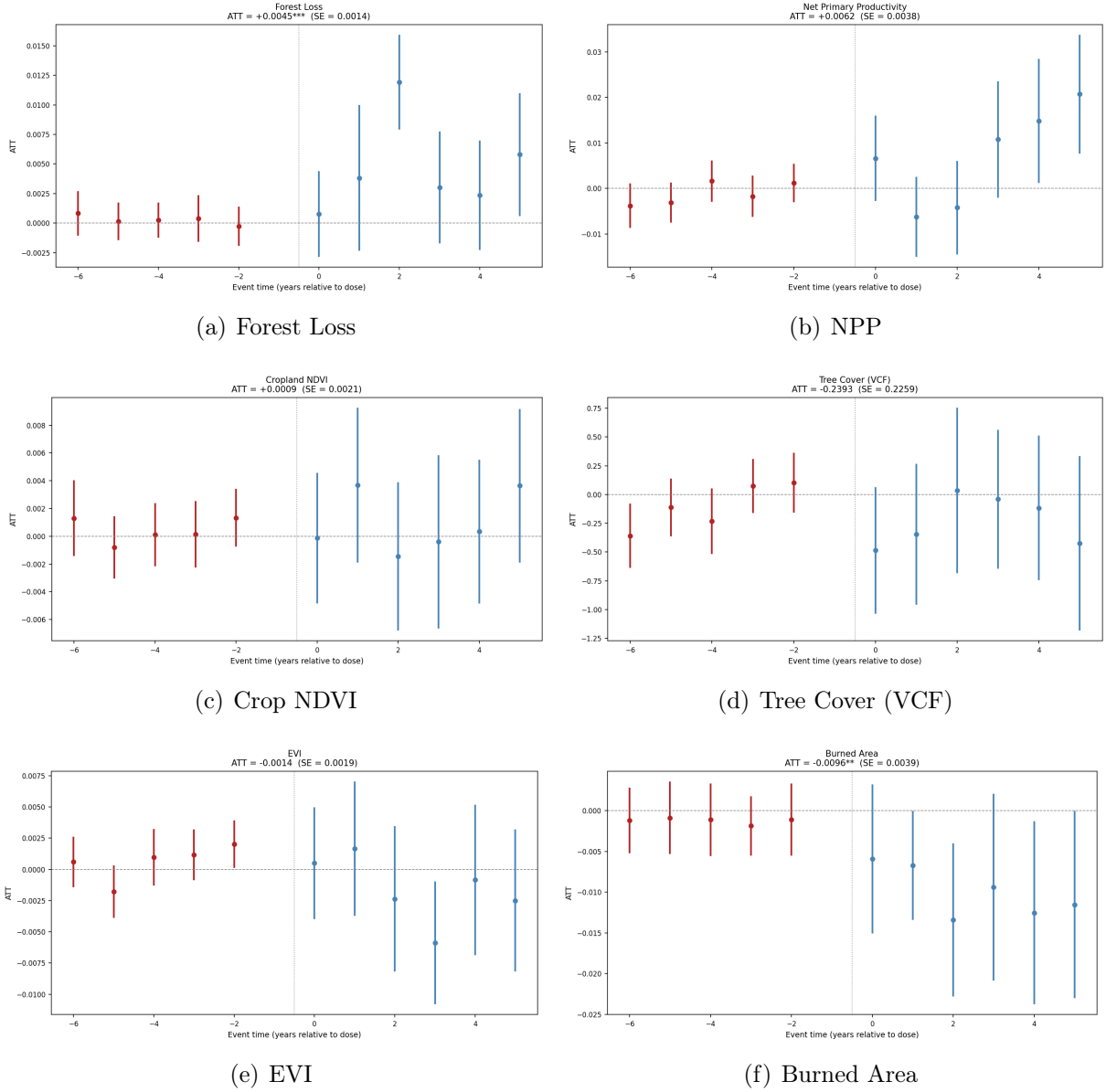


Figure 3: Event-study estimates pooling all admissible LSLT doses (woreda-years with ≥ 20 ha of new transfers), on the ever-treated sample: never-admissibly-dosed woredas are excluded, consistent with identification from investment timing conditional on being targeted. Each k -th-dose event is compared against woredas with the identical prior dose-year history that have not yet received a k -th dose (de Chaisemartin & D’Haultfœuille, 2024); the first-dose case reproduces the Callaway-Sant’Anna estimator on this sample, aggregated over the same horizon window. Overall ATTs (panel titles) average post-treatment blocks at event times 0–5, matching the displayed window. Whiskers show 95% CIs; all standard errors, per event-time bin and overall (panel titles), are finite-population AAIW interpolations of design and woreda-clustered variances. See Table 3 for effects by dose number and Table 18 for alternative control pools.

Table 2: Effect of LSLT Doses on AgSS Agricultural Outcomes: Pooled and by Dose Number

	Pooled	1st dose	2nd dose	3rd+ dose
Yield (kg/ha)	−23.1 (78.6)	+93.2 (153.7)	−65.8 (105.7)	−289.2*** (56.2)
Area (log ha)	−0.022 (0.079)	+0.021 (0.129)	−0.144 (0.130)	+0.157*** (0.045)
Fertilizer	−0.0110 (0.0090)	−0.0105 (0.0171)	−0.0209* (0.0126)	+0.0128* (0.0078)
Improved Seed	−0.0008 (0.0020)	−0.0036 (0.0041)	+0.0028 (0.0024)	−0.0022 (0.0023)
Irrigation	−0.0088*** (0.0030)	−0.0061 (0.0061)	−0.0076** (0.0032)	−0.0195** (0.0081)
Crop Damage	−0.0014 (0.0156)	−0.0089 (0.0220)	−0.0164 (0.0240)	+0.0596*** (0.0146)
Switcher events	121	57	45	19

Notes: CS-style aggregation of clean 2×2 blocks pooling all admissible LSLT doses (a dose is a woreda-year with ≥ 20 ha of new MOA transfers). The k -th-dose effect compares k -th-dose switchers to woredas with the identical prior dose-year history that have not yet received a k -th dose (de Chaisemartin–D’Haultfœuille 2024). The sample drops never-admissibly-dosed woredas, consistent with identification from investment timing conditional on being targeted; the first-dose column reproduces the Callaway–Sant’Anna estimator with not-yet-treated controls on this sample, aggregated over the same horizon window. Overall ATTs average post-treatment blocks at event times 0–5 (the plotted window) with switcher-count weights. Standard errors follow the finite-population variance of Abadie, Athey, Imbens, and Wooldridge (2020): a design component (stratified Neyman within dose-history cells) and a sampling component (woreda-clustered) weighted by the sampling share $\rho = n/N$, the estimation sample’s share of the full woreda universe. Pure design and clustered SEs (the $\rho = 1$ and $\rho = 0$ poles) are reported in the replication output. The AgSS panel is an unbalanced repeated cross-section; each block requires a woreda only at its two endpoint years. 131 of 362 switcher events have same-history controls with the required survey coverage. The 3rd+ column pools adjacent dose numbers whose separate estimates are mechanically negatively correlated (shared years switch base/endpoint roles). With 19 events, largely in single-switcher cells where the treated-arm variance (including treatment-effect heterogeneity) is unidentified under any variance estimator, reported precision in that column is a lower bound on uncertainty; the design-based SE is the more conservative of the available options. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 18 probes the pooled estimates across control-pool definitions: coarsened history matching, retaining never-treated woredas as first-dose controls, restricting non-switcher controls to woredas with positive MOA-designated potential investment area, requiring controls to never receive their k -th dose, and weighting blocks by dose hectares. The burned-area effect is significant in every specification and the forest-loss effect is stable in sign but attenuated under coarse history matching and hectare weights; the NPP effect is sensitive to the control pool, shrinking when controls are restricted to comparable (targeted or PIA-eligible) woredas — one reason we treat the productivity result as suggestive. The hectare-weighted column is a per-hectare estimand whose NPP and cropland-NDVI cells fail pre-treatment placebos and should not be read causally (table notes). Section 7.5 reports the first-treatment CS battery (full-sample, PIA, never-treated, BJS imputation, TWFE), whose $k = 1$ estimates the pooled design nests.

Table 3: Effect of LSLT Doses on Remote-Sensing Outcomes: Pooled and by Dose Number

	Pooled	1st dose	2nd dose	3rd+ dose
Net Primary Productivity	+0.0062 (0.0038)	+0.0019 (0.0058)	+0.0039 (0.0061)	+0.0237*** (0.0044)
Cropland NDVI	+0.0009 (0.0021)	+0.0027 (0.0031)	−0.0037 (0.0036)	+0.0065** (0.0026)
Forest Loss	+0.0045*** (0.0014)	+0.0083*** (0.0022)	+0.0009 (0.0026)	+0.0011 (0.0022)
Tree Cover (VCF)	−0.2393 (0.2259)	−0.1563 (0.3187)	+0.0753 (0.3977)	−1.1739*** (0.3188)
EVI	−0.0014 (0.0019)	−0.0047* (0.0026)	−0.0025 (0.0032)	+0.0109*** (0.0022)
Evapotranspiration	−2.15 (3.00)	−6.19 (4.51)	−7.58* (4.55)	+21.67*** (3.53)
Burned Area	−0.0096** (0.0039)	−0.0070 (0.0067)	−0.0105* (0.0056)	−0.0155*** (0.0044)
Forest Share	+0.0036*** (0.0014)	+0.0053*** (0.0019)	+0.0012 (0.0019)	+0.0042*** (0.0010)
Cropland Share	−0.0063** (0.0030)	−0.0046 (0.0058)	−0.0134*** (0.0040)	+0.0046** (0.0019)
Grassland Share	+0.0053 (0.0038)	+0.0044 (0.0059)	+0.0069 (0.0058)	+0.0043 (0.0038)
Savanna Share	−0.0017 (0.0032)	−0.0040 (0.0052)	+0.0064 (0.0044)	−0.0131*** (0.0029)
Switcher events	215	105	74	36

Notes: CS-style aggregation of clean 2×2 blocks pooling all admissible LSLT doses (a dose is a worda-year with ≥ 20 ha of new MOA transfers). The k -th-dose effect compares k -th-dose switchers to wordas with the identical prior dose-year history that have not yet received a k -th dose (de Chaisemartin–D’Haultfœuille 2024). The sample drops never-admissibly-dosed wordas, consistent with identification from investment timing conditional on being targeted; the first-dose column reproduces the Callaway–Sant’Anna estimator with not-yet-treated controls on this sample, aggregated over the same horizon window. Overall ATTs average post-treatment blocks at event times 0–5 (the plotted window) with switcher-count weights. Standard errors follow the finite-population variance of Abadie, Athey, Imbens, and Wooldridge (2020): a design component (stratified Neyman within dose-history cells) and a sampling component (worda-clustered) weighted by the sampling share $\rho = n/N$, the estimation sample’s share of the full worda universe. Pure design and clustered SEs (the $\rho = 1$ and $\rho = 0$ poles) are reported in the replication output. Exact history matching drops switcher events without same-history controls (152 of 378, concentrated at $k \geq 4$). Among the six headline outcomes, joint pre-treatment placebo tests pass except for Tree Cover and EVI ($p = 0.01$); several secondary rows (evapotranspiration, land-cover shares) also fail and are reported for completeness only. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.2 Gambela Pixel-Level Results

Here we present event-study estimates of the treatment effects on cells cropped at baseline in the immediate vicinity of LSLT concessions in Gambela. In particular, we estimate effects on remote-sensed outcomes for baseline-cropped pixels within a 5km buffer of the LSLT boundary. In contrast to the national-level results, we find significant decreases in NPP, NDVI, and EVI in Gambela, though the vegetation-index rows fail joint pre-treatment placebo tests and warrant caution; the burning, tree-cover, and land-cover results are placebo-clean. Likewise we find no effect on forest loss and a significant decrease in VCF. There are a number of explanations for these differences. First, Gambela is a small lowland region that has fundamentally different agricultural and demographic characteristics than highland Ethiopia. As we discuss in Section 5.2.1, only about 20% of LSLTs in Gambela engaged in cultivation, but simply took ownership of a large tract of land. In contrast, over 80% of LSLTs in Amhara undertook cultivation (Eisfelder et al., 2024).

5.2.1 Actual vs. Planned

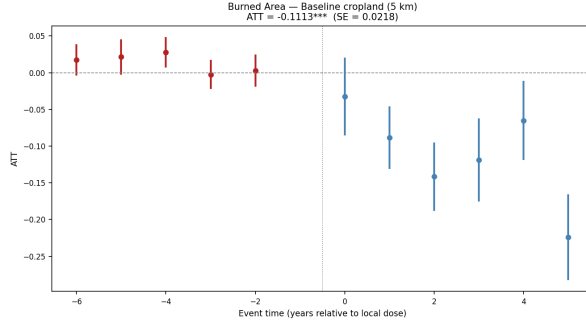
We now address to what extent the effects in Gambela—and their departure from those at the national level—are driven by LSLTs who actually entered vs. those who simply acquired land using the subsample of LSLTs analyzed by Eisfelder et al. (2024).

We find that where LSLTs actually engaged in cultivation, forest loss slows down in the surrounding buffers, but exhibits no effects where LSLTs did not. In contrast, canopy cover increases surrounding both LSLTs while the decrease in burned area is driven by areas in which LSLTs did not cultivate. The decrease in agricultural productivity is common to both cultivation and non-cultivation LSLTs. This leads to a number of conclusions. First, the overall effects from the subsample of Gambela in the Eisfelder et al. (2024) are different than those of the full region. Second, with this in mind, LSLTs that actually cultivate if anything seem to reduce forest loss, but they do not drive the negative spillovers in agricultural productivity.

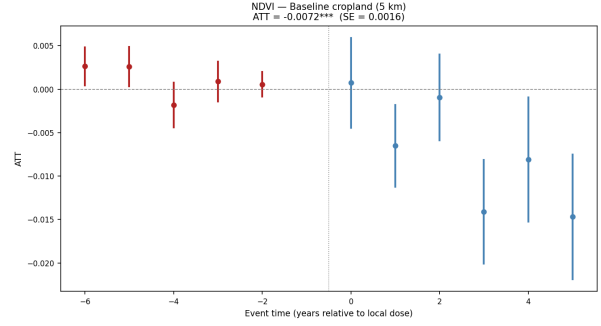
5.3 Gambela Survey

We estimate the causal effects of LSLT proximity on household welfare using Double-Debiased Machine Learning (DML) with a partially linear regression model. Treatment is measured as the total LSLT area (in 1,000 ha) within three distance rings around each household: 0–5 km, 5–10 km, and 10–20 km. XGBoost is used for nuisance estimation with 50 cross-fitting repetitions (median-aggregated), and continuous outcomes are winsorized at the 99th percentile; standard errors are clustered at the kebele level ($n = 103$ clusters). Results are stable across alternative nuisance learners (stacking ensemble, Random Forest, Elastic Net, post-double lasso; see robustness tables in the appendix).

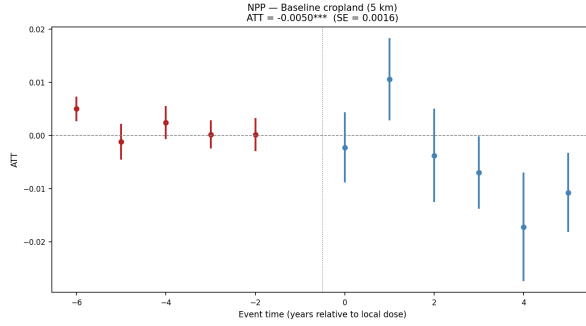
The only significant effect is an increase of 1.2 percentage points (off a base of 25%) in the share of households engaging in wage labor per 1,000 ha of LSLTs within 5-10km. There are no effects on agricultural, business, or total income, nor on household consumption at any distance from LSLTs. Again, these estimates are at a much longer time horizon from LSLT entry than the event-study estimates and one should bear in mind that the majority of LSLTs in Gambela were never operationalized.



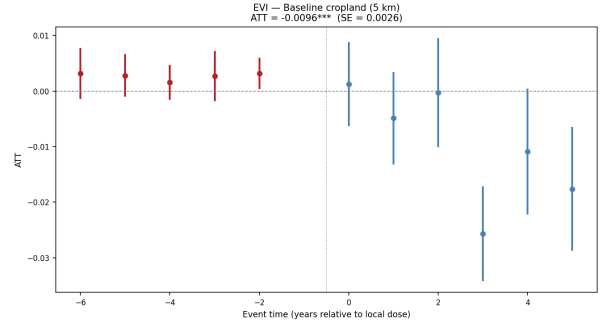
(a) Burned Area



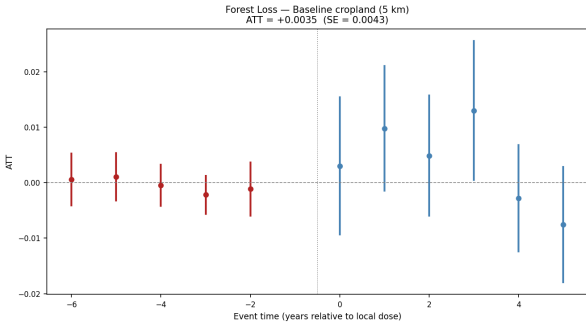
(b) NDVI



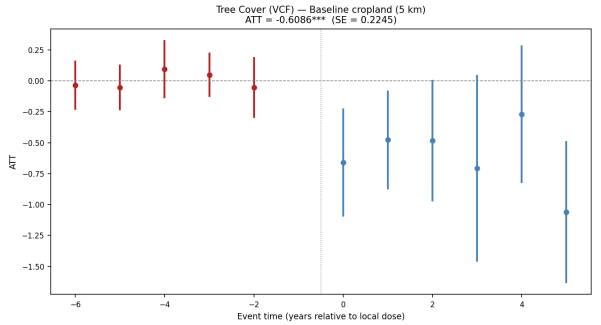
(c) NPP



(d) EVI

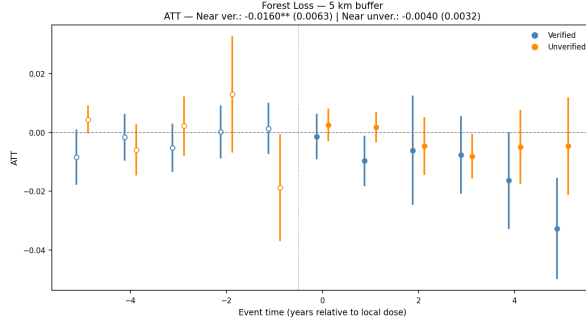


(e) Forest Loss

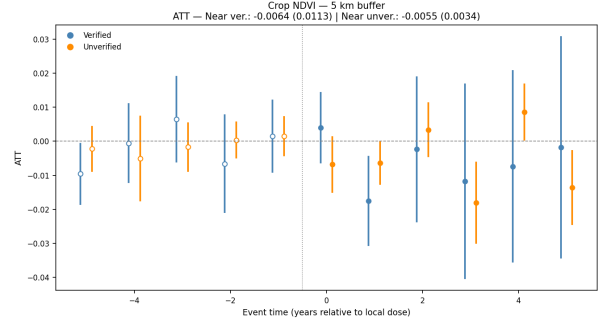


(f) Tree Cover (VCF)

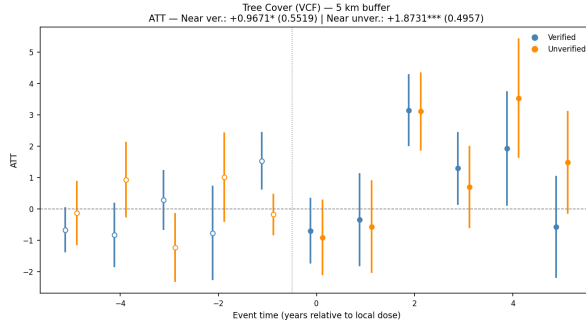
Figure 4: Pooled dose estimator at the pixel level, Gambela: baseline-cropped cells (MODIS cropland in 2001) within 5 km of a dated parcel, the main spillover specification. Each additional local dose (distinct parcel-entry year within 5 km) is compared against pixels with the identical prior local dose history (de Chaisemartin & D’Haultfoeuille, 2024); overall ATTs aggregate event times 0–5. Whiskers show 95% CIs from parcel-clustered SEs; overall ATT and design-based SE in panel titles. See Table 4 for on-parcel results.



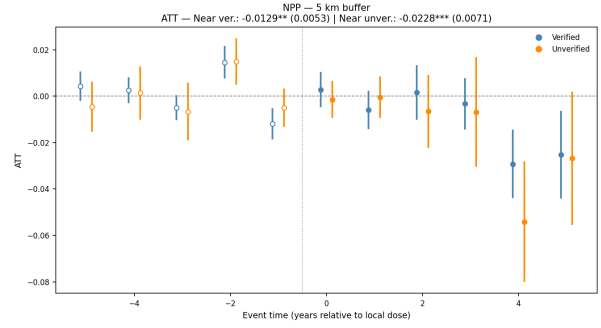
(a) Forest Loss



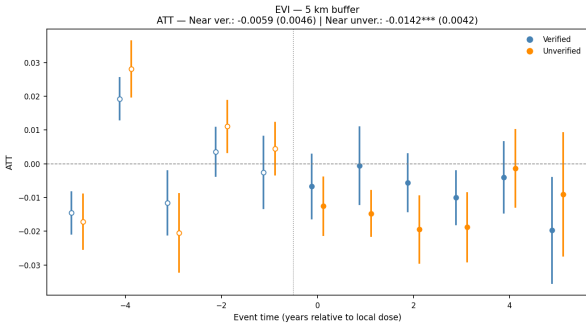
(b) Crop NDVI



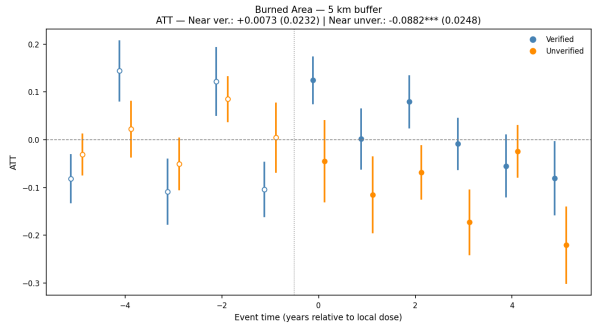
(c) Tree Cover (VCF)



(d) NPP



(e) EVI



(f) Burned Area

Figure 5: Buffer event-study estimates by GIZ field-verified production status: GEE outcomes, pixels within 5 km of LSLT parcels. Verified (blue) and unverified (orange) proximity plotted on common axes; open markers denote pre-treatment periods. Event time is relative to the first crosswalk-dated parcel entry within 5 km. Whiskers show 95% CIs; overall ATTs and SEs reported in panel titles. Standard errors clustered by parcel. Pre-period divergence in burned area near unverified parcels is consistent with anticipatory encroachment ahead of enclosure; we treat that pattern as suggestive.

Table 4: Pooled Dose Estimator at the Pixel Level: All Gambela

	On-parcel	Baseline cropland (5 km)
Burned Area	−0.1322*** (0.0077)	−0.1113*** (0.0218)
NDVI	−0.0027 (0.0020)	−0.0072*** (0.0016)
NPP	−0.0056*** (0.0006)	−0.0050*** (0.0016)
EVI	−0.0139*** (0.0008)	−0.0096*** (0.0026)
Forest Loss	−0.0002 (0.0013)	+0.0035 (0.0043)
Tree Cover (VCF)	−0.674*** (0.085)	−0.609*** (0.224)
Forest Share	−0.0047*** (0.0010)	+0.0008 (0.0019)
Cropland Share	−0.0041 (0.0029)	−0.0098 (0.0105)
Grassland Share	+0.0059* (0.0035)	+0.0546*** (0.0146)
Savanna Share	+0.0029 (0.0037)	−0.0455*** (0.0157)
Switching pixels	32, 660	4, 349

Notes: Pooled dose estimator over all crosswalk-dated LSLT parcels in Gambela (not restricted to the GIZ subsample). A pixel’s dose history is the sequence of distinct parcel-entry years within 5 km; each additional local dose is compared against pixels with the identical prior local history that have not yet received it, and clean 2×2 blocks are aggregated with switcher-count weights over event times 0–5. On-parcel: pixels inside parcel boundaries. Baseline cropland (the main spillover specification): cells with MODIS cropland in 2001 lying within 5 km of a dated parcel. On the cropland sample the vegetation-index rows fail joint pre-treatment placebo tests and should be read with caution; tree cover, land-cover shares, and forest loss pass. Design-based SEs (stratified Neyman within history cells, parcels as randomization units); the pixel panels cover their full universes, so the finite-population sampling share is one. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

To shed light on actual engagement with LSLTs Figure 7 and Figure 8 show descriptive evidence on self-reported household interactions with LSLTs by distance, corroborating the spatial gradient: engagement through wage labor is most common at intermediate distances, while direct land-use conflicts are extremely rare.

Together these results show that LSLTs had a minimal economic impact on households in Gambela. The following subsection discusses further whether this simply because few LSLTs entered, or whether even those that did enter had little effect on the local economy.

Table 5: Buffer Effects by GIZ Field-Verified Production Status (Gambella, 5 km buffers)

	Near verified	Near unverified	No GIZ ≤ 5 km
Cropland NDVI	−0.0064 (0.0113)	−0.0055 (0.0034)	+0.0142 (0.0118)
NPP	−0.0129** (0.0053)	−0.0228*** (0.0071)	+0.0095*** (0.0030)
Tree Cover (VCF)	+0.9671* (0.5519)	+1.8731*** (0.4957)	+0.9504** (0.3921)
Burned Area	+0.0073 (0.0232)	−0.0882*** (0.0248)	−0.1022*** (0.0093)
Forest Loss	−0.0160** (0.0063)	−0.0040 (0.0032)	+0.0106*** (0.0032)
Cropland Share	−0.0013 (0.0038)	−0.0123** (0.0052)	−0.0003*** (0.0001)
Forest Share	−0.0030 (0.0141)	+0.0146** (0.0068)	−0.0208*** (0.0062)
Grassland Share	−0.0068 (0.0085)	−0.0020 (0.0112)	−0.0072*** (0.0024)
Parcel clusters	256	132	171

Notes: Pixel-level Callaway–Sant’Anna overall ATTs (uniform cohort weights), standard errors clustered by parcel. Verified: pixel lies in / within 5 km of a GIZ RS4AIM (Eisfelder et al., 2024) LSAI parcel containing at least one 2021–22 ground-truth survey point (96% of on-parcel points record active cropland); Unverified: GIZ parcel never visited by surveyors. Verification is post-treatment and surveyor-selected, so columns are descriptive splits, not causal moderation. First-local-exposure design: event time is the earliest crosswalk-dated parcel entry within 5 km. Cells failing within-group joint placebo tests at 5% are reported in the replication output; effects near unverified parcels show pre-allocation divergence consistent with anticipatory encroachment and should be read as suggestive. Tree Cover estimated on the complete-series pixel subsample. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 6: DML Estimates: Wages and Employment

	Log wage income	Has wage income	Log business income
<i>0–5 km</i>	+0.008 (0.037)	+0.001 (0.012)	+0.014 (0.091)
<i>5–10 km</i>	−0.008 (0.020)	+0.012** (0.006)	+0.020 (0.034)
<i>10–20 km</i>	−0.005 (0.003)	+0.001 (0.002)	+0.000 (0.011)
Mean dep. var.	10.961	0.248	9.986
<i>N</i>	1,198	4,814	768

Table 7: DML Estimates: Agricultural Outcomes

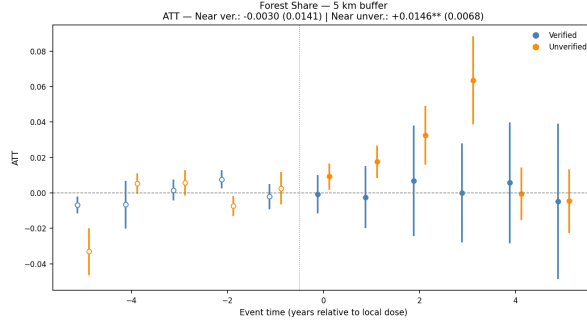
	Log crop income	Log harvest (kg)	# crops
<i>0–5 km</i>	−0.017 (0.053)	+0.004 (0.044)	+0.026 (0.056)
<i>5–10 km</i>	−0.033 (0.027)	+0.005 (0.017)	+0.043 (0.030)
<i>10–20 km</i>	+0.011 (0.008)	−0.000 (0.006)	+0.004 (0.008)
Mean dep. var.	9.252	7.167	3.070
<i>N</i>	1,940	3,797	3,819

Table 8: DML Estimates: Consumption and Total Income

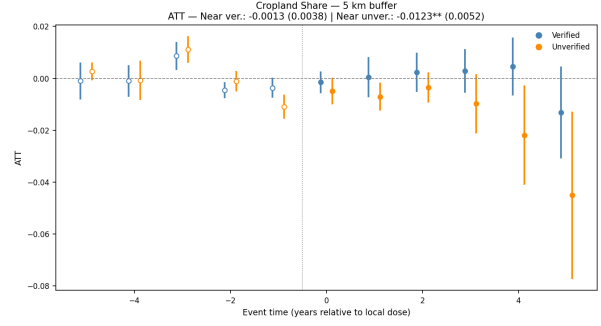
	Log consumption	Log total HH income
<i>0–5 km</i>	−0.010 (0.026)	+0.006 (0.043)
<i>5–10 km</i>	−0.005 (0.011)	+0.022 (0.016)
<i>10–20 km</i>	+0.005 (0.004)	+0.001 (0.006)
Mean dep. var.	7.754	10.745
<i>N</i>	4,814	4,052

Table 9: DML Estimates: Environmental Income

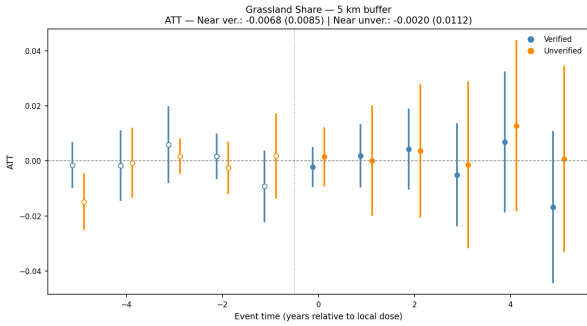
	Log env income	Has env income
<i>0–5 km</i>	+0.000 (0.066)	+0.008 (0.009)
<i>5–10 km</i>	+0.014 (0.038)	−0.003 (0.004)
<i>10–20 km</i>	−0.001 (0.013)	+0.001 (0.002)
Mean dep. var.	8.831	0.210
<i>N</i>	1,008	4,814



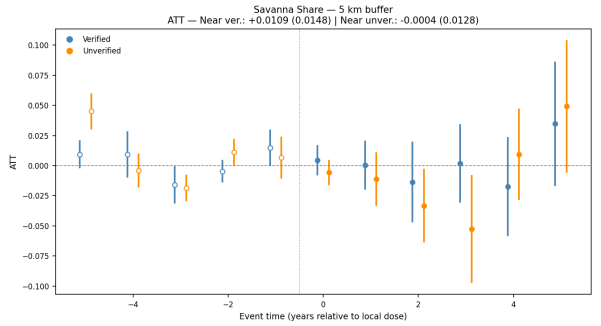
(a) Forest Share



(b) Cropland Share



(c) Grassland Share



(d) Savanna Share

Figure 6: Buffer event-study estimates by GIZ field-verified production status: land cover shares. Design as in Figure 5.

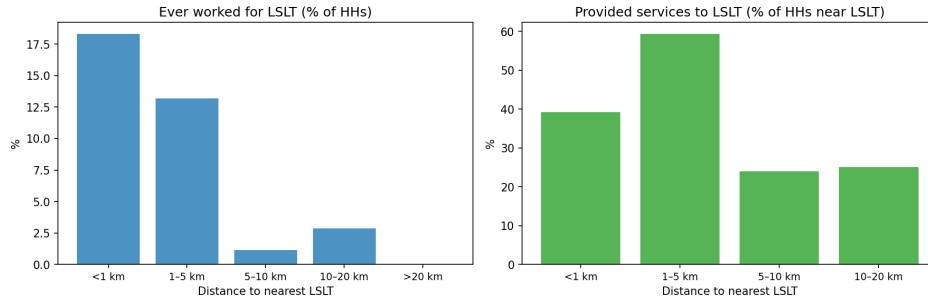


Figure 7: Household interactions with LSLT sites by distance ring. Shares are conditional on being within 20 km of at least one LSLT parcel. Source: Gambela household survey.

5.3.1 Actual vs. Planned

Comparing households surrounding cultivated vs. uncultivated parcels in the GIZ subsample in Table 10 confirms that the only effect of LSLT entry—on wage employment—is driven by actual investments. However, in this subsample it shows up as an increase in labor income conditional on wage work, rather than on the probability of wage employment. Again, this appears to be a compositional shift (offset by large but imprecisely estimated effects on crop income), as overall income and consumption do not increase. In contrast, there is suggestive

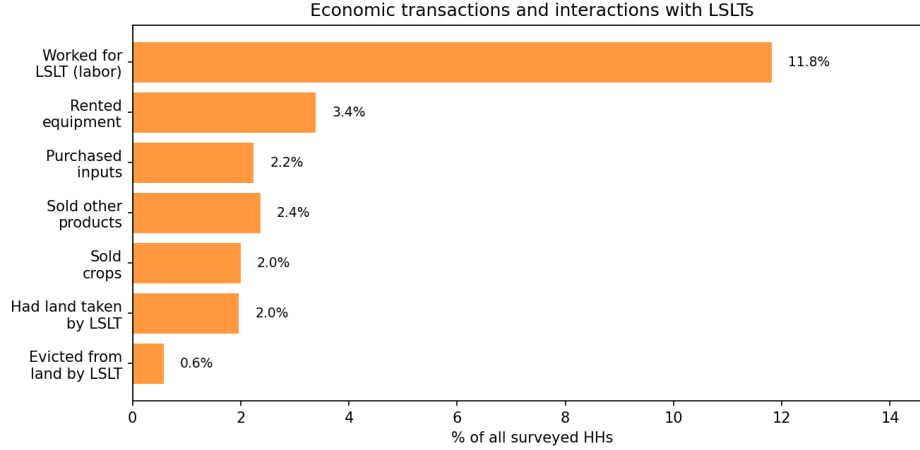


Figure 8: Household interactions with LSLT sites by type. Source: Gambela household survey.

evidence of greater engagement in agriculture (a suggestive increase in the share of HH with crop income in the 5-10km buffer and a positive but noisy increase in log crop income in the 0-5km buffer) around uncultivated parcels. Tables 23 and 24 confirm that the buffer zones around these parcels exhibit parallel trends tho those around uncultivated ones on key time-varying outcomes.

5.3.2 Robustness

Table 19–Table 22 replicate the key outcomes under alternative nuisance learners (stacking ensemble, Random Forest, Elastic Net) and post-double Lasso OLS. The 5–10 km wage-employment effect is positive across all alternative learners. The directionally negative consumption effect at 0–5 km reaches conventional significance only in the post-lasso benchmark, reflecting lower power from kebele-clustered variance estimation with 103 clusters under the DML specs.

6 Conclusion

This paper studies the effect of LSLTs on local economic and environmental outcomes in Ethiopia, by combining a national level event-study on remote sensed outcomes and an original household survey in Gambela. At the national level, we find suggestive evidence of a small increase in agricultural productivity but reduced use of irrigation. LSLTs appear to increase deforestation but exhibit a null effect on overall tree cover and reduce crop burning. In Gambela, where a large number of LSLTs never engaged in production, we again see minimal effects on the local economy, apart from a small increase in the share of households engaging in wage employment. Notably, the small effects on wage labor without any effects on total income or consumption occur even where LSLTs did engage in cultivation and therefore are not just an artifact of unrealized investments. Cultivated LSLTs also seem to have preserved tree cover. Taken together, these results show that LSLTs in Ethiopia were neither

Table 10: Household Outcomes and Exposure to Verified vs Unverified GIZ Investments

	Verified		Unverified		<i>N</i>
	0–5 km	5–10 km	0–5 km	5–10 km	
Log consumption	−0.093* (0.051)	+0.014 (0.021)	+0.018 (0.034)	+0.013 (0.013)	4814
Log total HH income	+0.047 (0.101)	−0.020 (0.058)	+0.015 (0.075)	+0.033 (0.037)	4052
Log crop income	−0.116 (0.151)	+0.017 (0.076)	+0.270 (0.179)	+0.005 (0.059)	1940
Log wage income	+0.159* (0.089)	−0.005 (0.038)	−0.035 (0.100)	−0.012 (0.035)	1198
Has crop income	+0.002 (0.039)	+0.011 (0.019)	+0.000 (0.040)	+0.019 (0.013)	3776
Has wage income	−0.006 (0.039)	−0.002 (0.017)	−0.006 (0.043)	−0.006 (0.012)	4814

Notes: Double/debiased ML (partially linear model, XGBoost nuisances, 3-fold cross-fitting, 50 repetitions (median-aggregated)), standard errors clustered by kebele. All four exposures enter jointly. Exposure = GIZ LSAI parcel area (1,000 ha) within the 0–5 km disc / 5–10 km annulus around the household, split by ground-truth verification status. Continuous outcomes are winsorized at the 99th percentile; log outcomes drop zeros (intensive margin) and are paired with extensive-margin dummies. Controls: the full baseline observable set (soil chemistry, terrain, access, potential investment area). Among exposed households the verified share of nearby GIZ area is largely predictable from these observables ($R^2 = 0.91$), so estimates are conditional-on-observables, mirroring the main design.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

the economic boon proponents hoped for nor the catastrophe critics warned about and suggest that other rural development mechanisms may warrant attention instead.

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7 Appendix

7.1 Additional Tables and Figures

Table 11: Summary of quantitative LSLT impact studies

Paper	Context	Method	Key finding
Economic outcomes			
Ali et al. (2019)	Ethiopia	PSM, cross-section	No job creation; modest tech adoption
Shete and Rutten (2015)	Ethiopia (Oromia)	PSM, cross-section	Negative food security, income
Deininger and Xia (2016)	Mozambique	PSM	Positive inputs; negative well-being
Bottazzi et al. (2018)	Sierra Leone	PSM	Higher income; heterog. by gender
Anti (2021)	Cambodia	TWFE	Negative consumption; wage shift
Jiao et al. (2015)	Cambodia	Matching	Negative income, land, livestock
Bunte et al. (2018)	Liberia	TWFE	No growth effects
Müller et al. (2021)	Multi-country	Meta-analysis	Increased production, food insecurity
Akalu et al. (2025)	Multi-country	Meta-analysis (143 studies)	Modest positive overall; heterog. by country
Sy and Gehrke (2026)	Senegal	TWFE	Increased consumption and wage labor

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Paper	Context	Method	Key finding
Environmental outcomes			
Davis et al. (2015)	Cambodia	Matching	Substantial forest loss
Davis et al. (2020)	15 countries	Matching	Forest loss; heterogeneous
Rulli et al. (2019)	Indonesia	Matching	Forest loss, fragmentation
Liao et al. (2020)	Eth., Cambodia, Peru, Liberia	Simulation	Projected large forest loss
Social outcomes			
Wegenast et al. (2022)	Multi-country	Survey	Reduced trust
Khadjavi et al. (2021)	Multi-country	Lab-in-field	Increased trust/social capital
Hajjar et al. (2020)	Ethiopia	Qualitative (4 farms)	Land use change, labor reallocation

Table 12: Summary Statistics — National Woreda Panel

	<i>N</i>	Mean	SD	Min	Max
<i>Panel A: GEE Remote Sensing Outcomes</i>					
NPP (kgC/m ² /yr)	15,755	0.834	0.421	0.000	2.133
Crop NDVI	14,191	0.752	0.073	0.233	0.922
Raw NDVI	15,847	0.695	0.160	0.090	0.922
Forest Loss (fraction)	15,847	0.009	0.019	0.000	0.369
Tree Cover (VCF %)	15,847	13.592	11.858	0.000	77.739
EVI	15,847	0.477	0.134	0.048	0.715
Burned Area (fraction)	15,847	0.036	0.103	0.000	0.913
<i>Panel B: AgSS Agricultural Outcomes</i>					
Yield (kg/ha)	8,044	2,306	2,011	0.000	9,983
Fertilizer Use (share)	9,212	0.363	0.210	0.000	0.907
Improved Seed (share)	9,212	0.027	0.044	0.000	0.542
Irrigation (share)	9,212	0.023	0.059	0.000	0.762
Crop Damage (share)	9,022	0.434	0.173	0.000	5.160
Mean Plot Area (ha)	8,929	0.109	0.082	0.000	2.323
GEE outcomes cover 689 woredas, 2001–2023. AgSS outcomes cover 905 woredas, 2003–2020.					

7.2 Actual vs. Planned Balance Checks

7.3 Kebele-Level Results (Gambela)

We replicate the event study at the more granular kebele level in Gambela region, where we have geolocated LSLT boundaries matched to individual kebeles. This yields 36 treated kebeles (treatment years 2007–2014) and 162 never-treated kebeles. We estimate CS with both ever-treated-only (not-yet-treated controls) and full-sample (never-treated controls) specifications, clustering standard errors at the kebele level.

7.4 On-Parcel Results (GIZ Subsample)

Table 13: Summary Statistics — Gambella Kebele Panel

	<i>N</i>	Mean	SD	Min	Max
<i>Panel A: GEE Remote Sensing Outcomes</i>					
NPP (kgC/m ² /yr)	4,554	0.612	0.399	0.000	1.777
Crop NDVI	3,128	0.808	0.036	0.589	0.906
Raw NDVI	4,554	0.824	0.048	0.593	0.936
Forest Loss (fraction)	4,554	0.011	0.029	0.000	0.607
Tree Cover (VCF %)	4,554	25.321	16.517	2.837	79.057
EVI	4,554	0.608	0.063	0.363	0.782
Burned Area (fraction)	4,554	0.230	0.245	0.000	1.000
<i>Panel B: Land Cover Shares</i>					
Forest Share	4,554	0.171	0.325	0.000	1.000
Cropland Share	4,554	0.028	0.062	0.000	0.633
Savanna Share	4,554	0.502	0.356	0.000	1.000
Grassland Share	4,554	0.264	0.289	0.000	1.000

198 kebeles in Gambella region, 2001–2023. Land cover shares from MODIS MCD12Q1.

Table 14: Summary Statistics — Gambella Household Survey

	<i>N</i>	Mean	SD	Min	Max
<i>Panel A: Key Outcomes</i>					
Log Consumption	4,833	7.754	0.690	1.204	11.578
Log Total HH Income	4,833	9.035	4.155	0.000	15.646
Log Wage Income	4,833	2.722	4.753	0.000	13.541
Log Crop Income	4,833	3.594	4.874	-12.172	15.201
Log Harvest (kg)	4,833	5.658	3.115	0.000	14.457
Log Crop Revenue	4,833	4.008	4.807	0.000	15.202
<i>Panel B: LSLT Exposure Measures</i>					
LSLT Area within 5 km (ha)	4,833	0.168	0.195	0.000	0.810
LSLT Count within 5 km	4,833	5.042	6.399	0.000	32.000
LSLT Area within 10 km (ha)	4,833	0.219	0.208	0.000	0.788
LSLT Count within 10 km	4,833	16.246	18.828	0.000	79.000
Min. Distance to LSLT (km)	4,814	0.411	0.537	0.000	3.645
Investor Share of Kebele Area	4,833	0.217	0.268	0.000	0.883

2,527 households across 2 survey rounds. Distance converted from decimal degrees ($1^\circ \approx 111$ km).

Table 15: Effect of LSLT Entry on Remote-Sensing Outcomes

	CS	CS (weighted)	TWFE	BJS
NPP	+0.0096 (0.0069)	+0.0289*** (0.0085)	+0.0304*** (0.0054)	+0.0084 (0.0061)
Cropland NDVI	+0.0000 (0.0035)	−0.0005 (0.0073)	+0.0016 (0.0016)	+0.0028 (0.0019)
NDVI (raw)	−0.0056 (0.0040)		−0.0005 (0.0015)	−0.0023 (0.0018)
Forest Loss	+0.0021 (0.0021)	+0.0017 (0.0030)	−0.0004 (0.0009)	+0.0057*** (0.0016)
Tree Cover (VCF)	+0.5026 (0.3200)	+0.4714 (0.6766)	+0.2593 (0.2046)	−0.0769 (0.2771)
EVI	−0.0089** (0.0036)		+0.0003 (0.0016)	−0.0042** (0.0019)
Burned Area	−0.0125 (0.0078)		−0.0180*** (0.0038)	−0.0126*** (0.0040)
Joint pre-trend p	0.051–0.995	0.006–0.672†	—	0.067–0.859

Notes: Each cell reports the overall ATT (standard error) for the indicated outcome–estimator pair. CS: Callaway-Sant’Anna with varying base period, not-yet-treated controls, ever-treated woredas only ($N = 102$); CHIRPS precipitation as covariate. CS (weighted): CS with first-entry area weights (R did package). TWFE: two-way fixed effects (entity + year), full sample. BJS: Borusyak-Jaravel-Spiess imputation estimator, ever-treated only. Standard errors clustered by woreda. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 16: Effect of LSLT Entry on Agricultural Outcomes (AgSS)

	CS	CS (weighted)	TWFE	BJS
Yield (kg/ha)	+181.4 (211.8)	−69.2 (269.6)	+388.4*** (115.9)	+193.1 (146.6)
Area (log ha)	+0.216 (0.166)	−0.325 (0.258)		+0.196** (0.074)
Fertilizer	−0.0133 (0.0293)	+0.0434 (0.0422)	−0.0072 (0.0081)	+0.0035 (0.0093)
Improved Seed	+0.0069 (0.0062)	+0.0006 (0.0037)	+0.0040 (0.0028)	+0.0046* (0.0024)
Irrigation	+0.0012 (0.0089)	+0.0197* (0.0106)	+0.0062** (0.0029)	−0.0010 (0.0035)
Crop Damage	−0.0529** (0.0255)	−0.0099 (0.0294)	+0.0089 (0.0129)	+0.0077 (0.0143)

Notes: Same estimator specifications as Table 15. Outcomes from the Agricultural Sample Survey (AgSS) repeated cross-section at the woreda level. Standard errors clustered by woreda. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 17: Robustness: CS Under Alternative Samples and Controls (GEE Outcomes)

	CS full sample	CS PIA controls	CS 2nd-entry
NPP	+0.0269*** (0.0054)	−0.0056 (0.0102)	+0.0141 (0.0325)
Cropland NDVI	−0.0027 (0.0028)	−0.0028 (0.0035)	−0.0031 (0.0082)
NDVI (raw)	−0.0046* (0.0026)	−0.0034 (0.0029)	+0.0125 (0.0083)
Forest Loss	+0.0014 (0.0012)	+0.0044** (0.0020)	−0.0067** (0.0030)
Tree Cover (VCF)	+0.4876* (0.2755)	+0.5007 (0.3430)	−5.6710*** (1.6798)
EVI	−0.0048** (0.0024)	−0.0077** (0.0036)	+0.0131 (0.0102)
Burned Area	−0.0254*** (0.0069)	−0.0092 (0.0102)	−0.0059 (0.0173)
<i>Joint pre-trend p (Wald χ^2)</i>			
NPP	0.764	0.052	0.687
Cropland NDVI	0.048†	0.630	0.025†
NDVI (raw)	0.301	0.712	0.180
Forest Loss	0.646	0.327	0.086
Tree Cover (VCF)	0.790	0.628	0.772
EVI	0.164	0.591	0.106
Burned Area	0.920	0.693	0.738

Notes: Callaway-Sant’Anna estimator with varying base period. Full sample: all wordas including never-treated as controls. PIA controls: ever-treated plus never-treated wordas with positive Potential Investment Area. 2nd-entry: treatment redefined as 2nd cumulative threshold crossing; single-entry wordas as controls. CHIRPS precipitation as covariate. Standard errors clustered by worda. †: joint pre-trend $p < 0.05$. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 18: Pooled Dose Estimator: Robustness to Sample and Control-Pool Definitions

	Baseline	Coarse hist.	Full sample	PIA controls	Never controls	Ha-weighted
Forest Loss	+0.0045*** (0.0014)	+0.0009 (0.0008)	+0.0018* (0.0010)	+0.0020 (0.0013)	+0.0024** (0.0010)	+0.0003 (0.0017)
Net Primary Productivity	+0.0062 (0.0038)	+0.0025 (0.0026)	+0.0136*** (0.0023)	+0.0073** (0.0035)	+0.0147*** (0.0023)	+0.0124** (0.0047)
Cropland NDVI	+0.0009 (0.0021)	+0.0005 (0.0013)	−0.0003 (0.0012)	+0.0002 (0.0019)	−0.0003 (0.0014)	+0.0074 (0.0054)
Tree Cover (VCF)	−0.239 (0.226)	−0.090 (0.154)	−0.040 (0.143)	+0.029 (0.199)	+0.091 (0.161)	−0.543* (0.279)
EVI	−0.0014 (0.0019)	−0.0007 (0.0012)	−0.0013 (0.0012)	−0.0015 (0.0018)	−0.0012 (0.0013)	+0.0021 (0.0038)
Burned Area	−0.0096** (0.0039)	−0.0044* (0.0024)	−0.0119*** (0.0021)	−0.0086** (0.0036)	−0.0125*** (0.0023)	−0.0072* (0.0042)
Switching woredas	105	105	106	106	106	105
Sampling share ρ	0.16	0.16	0.98	0.20	0.98	0.16

Notes: Overall pooled ATTs (all admissible doses, switcher-count weights) under alternative samples and control pools; finite-population AAIW SEs in parentheses, interpolating design and clustered variances with each spec’s sampling share ρ (bottom row; $N = 689$ woredas). *Baseline* (main spec): ever-treated sample — never-admissibly-dosed woredas dropped, so first-dose controls are future admissible recipients; controls are same-history woredas not yet at their k -th admissible dose. *Coarse hist.*: histories matched on (number of prior doses, cumulative-hectares bin) instead of exact dose-year sequences. *Full sample*: never-admissibly-dosed woredas retained as first-dose controls; repeated-dose ($k \geq 2$) estimates are unaffected by this choice by construction. *PIA controls*: non-switcher controls restricted to woredas with positive potential investment area. *Never controls*: controls must never receive their k -th dose within the sample period. *Ha-weighted*: ever-treated sample with blocks weighted by the switchers’ k -th-dose hectares rather than switcher counts (a per-hectare estimand; the top 20 of 459 dose events carry 45% of the weight). Caveat: under hectare weights the NPP and Cropland NDVI columns fail joint pre-treatment placebo tests ($p = 0.014$, $p = 0.040$) with pre-period drift of the same sign as the estimates — consistent with large transfers selecting sites on improving agricultural trajectories — so those two cells should not be given a causal interpretation; Tree Cover and Burned Area pass ($p = 0.15$, $p = 0.37$). *Any dose*: admissibility threshold lowered from 20 ha to any positive transfer. Micro-transfers both dilute the switcher-weighted average with near-zero-footprint events and shift woredas’ dose clocks to years with negligible land-scale activity, so attenuation of the land-cover effects toward zero is expected under this definition; burned area is invariant. *Dense forest*: ever-treated woredas whose 2001 MODIS dense-forest share exceeds 1% (52 of 107) — the forest-frontier subsample, where the deforestation effect roughly doubles against a baseline loss rate of 1.3% per year. The full-sample estimate is not measurement-diluted: restricting instead to woredas within Hansen’s measurement domain (2001 woody cover above 10%, 82 woredas) leaves it essentially unchanged (+0.0042, $p < 0.05$). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

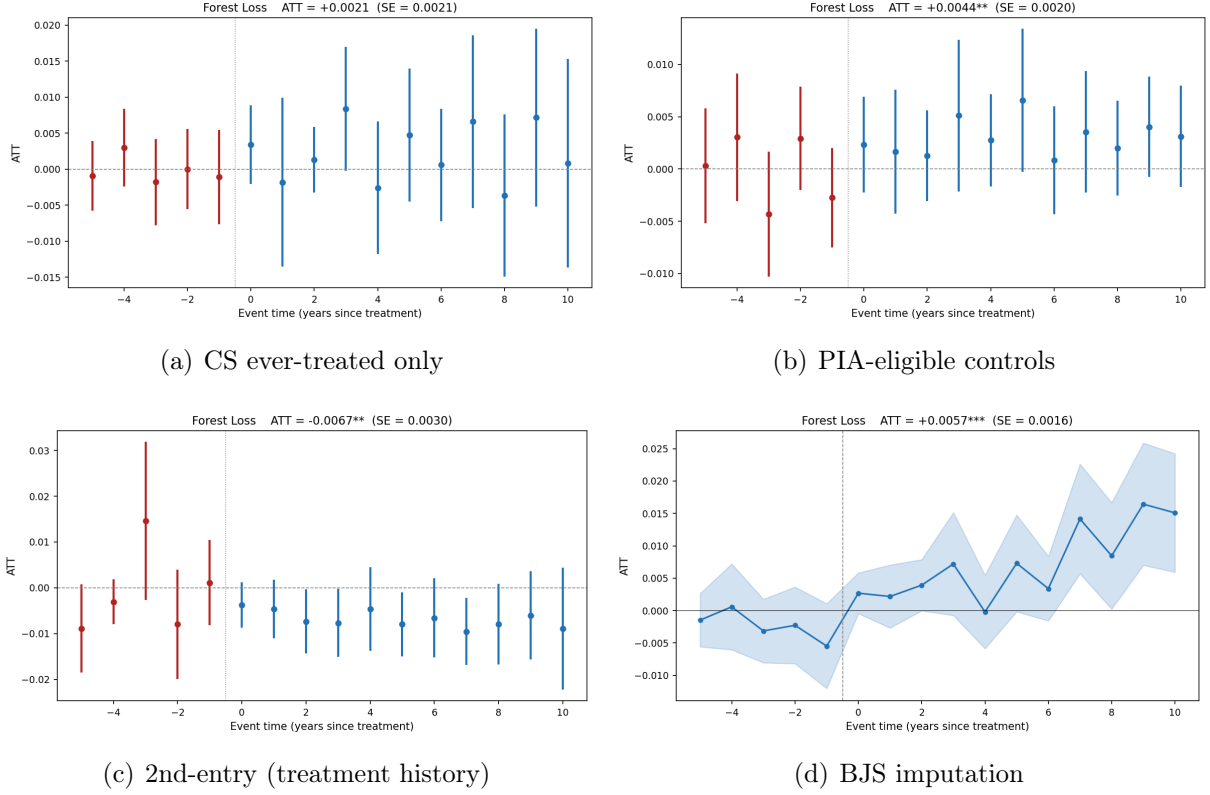


Figure 9: Forest Loss: Robustness across alternative specifications. Panel (a) restricts to ever-treated wordas with not-yet-treated controls. Panel (b) restricts never-treated controls to wordas with positive Potential Investment Area. Panel (c) redefines treatment as the second entry, conditioning on prior treatment history. Panel (d) uses the BJS imputation estimator for comparison.

Table 19: Robustness: Stacking Ensemble

	Log wage income	Has wage	Log crop income	Log harvest	Log cons.	Log HH income
<i>0–5 km</i>	+0.010 (0.128)	−0.009 (0.014)	+0.078 (0.166)	−0.128 (0.115)	−0.032 (0.026)	+0.072 (0.111)
<i>5–10 km</i>	+0.137** (0.061)	+0.009* (0.005)	+0.083 (0.058)	+0.041 (0.046)	−0.013 (0.010)	+0.086 (0.064)
<i>10–20 km</i>	+0.005 (0.019)	+0.000 (0.002)	−0.007 (0.020)	−0.008 (0.012)	+0.004 (0.004)	−0.025 (0.019)
<i>N</i>	4,814	4,814	4,814	4,814	4,814	4,814

Notes: Coefficients from Double ML-PLR with kebele-clustered standard errors in parentheses. Treatment variables measure LSLT area within each distance ring (1,000 ha). Controls: soil properties, terrain, climate, vegetation, road access, potential investment area, safety index, ethnicity indicators, and survey wave. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Nuisance learner: Stacking ensemble (XGBoost + RF + Elastic Net → Ridge meta-learner, $n_{\text{rep}} = 5$, $k = 3$ folds).

Table 20: Robustness: Random Forest

	Log wage income	Has wage	Log crop income	Log harvest	Log cons.	Log HH inc
<i>0–5 km</i>	+0.030 (0.048)	−0.004 (0.017)	+0.008 (0.062)	+0.055 (0.052)	−0.031 (0.028)	−0.116 (0.087)
<i>5–10 km</i>	−0.002 (0.019)	+0.010* (0.005)	−0.008 (0.028)	+0.006 (0.019)	+0.001 (0.012)	+0.015 (0.025)
<i>10–20 km</i>	−0.008 (0.008)	+0.000 (0.003)	+0.010 (0.010)	+0.006 (0.008)	+0.005 (0.004)	−0.007 (0.009)
Mean dep. var.	10.961	0.248	9.252	7.167	7.754	10.745
<i>N</i>	1,198	4,814	1,940	3,797	4,814	4,052

Table 21: Robustness: Elastic Net

	Log wage income	Has wage	Log crop income	Log harvest	Log cons.	Log HH inc
<i>0–5 km</i>	+0.004 (0.036)	+0.006 (0.013)	−0.025 (0.056)	+0.050 (0.036)	−0.007 (0.022)	+0.041 (0.047)
<i>5–10 km</i>	−0.002 (0.016)	+0.015*** (0.005)	−0.033 (0.024)	−0.009 (0.017)	−0.006 (0.010)	+0.017 (0.016)
<i>10–20 km</i>	−0.008 (0.005)	+0.002 (0.002)	+0.005 (0.006)	+0.000 (0.005)	+0.001 (0.002)	+0.001 (0.003)
Mean dep. var.	10.961	0.248	9.252	7.167	7.754	10.745
<i>N</i>	1,198	4,814	1,940	3,797	4,814	4,052

Table 22: Robustness: Post-Double Lasso OLS

	Log wage income	Has wage	Log crop income	Log harvest	Log cons.	Log HH inc
<i>0–5 km</i>	+0.010 (0.054)	−0.026 (0.016)	+0.001 (0.064)	+0.039 (0.041)	−0.044* (0.024)	−0.023 (0.057)
<i>5–10 km</i>	+0.002 (0.026)	+0.021*** (0.007)	−0.047* (0.028)	−0.015 (0.021)	+0.007 (0.013)	+0.030 (0.022)
<i>10–20 km</i>	−0.006 (0.006)	−0.002 (0.002)	+0.011 (0.009)	+0.004 (0.007)	+0.002 (0.003)	−0.006 (0.006)
Mean dep. var.	10.961	0.248	9.252	7.167	7.754	10.745
<i>N</i>	1,198	4,814	1,940	3,797	4,814	4,052

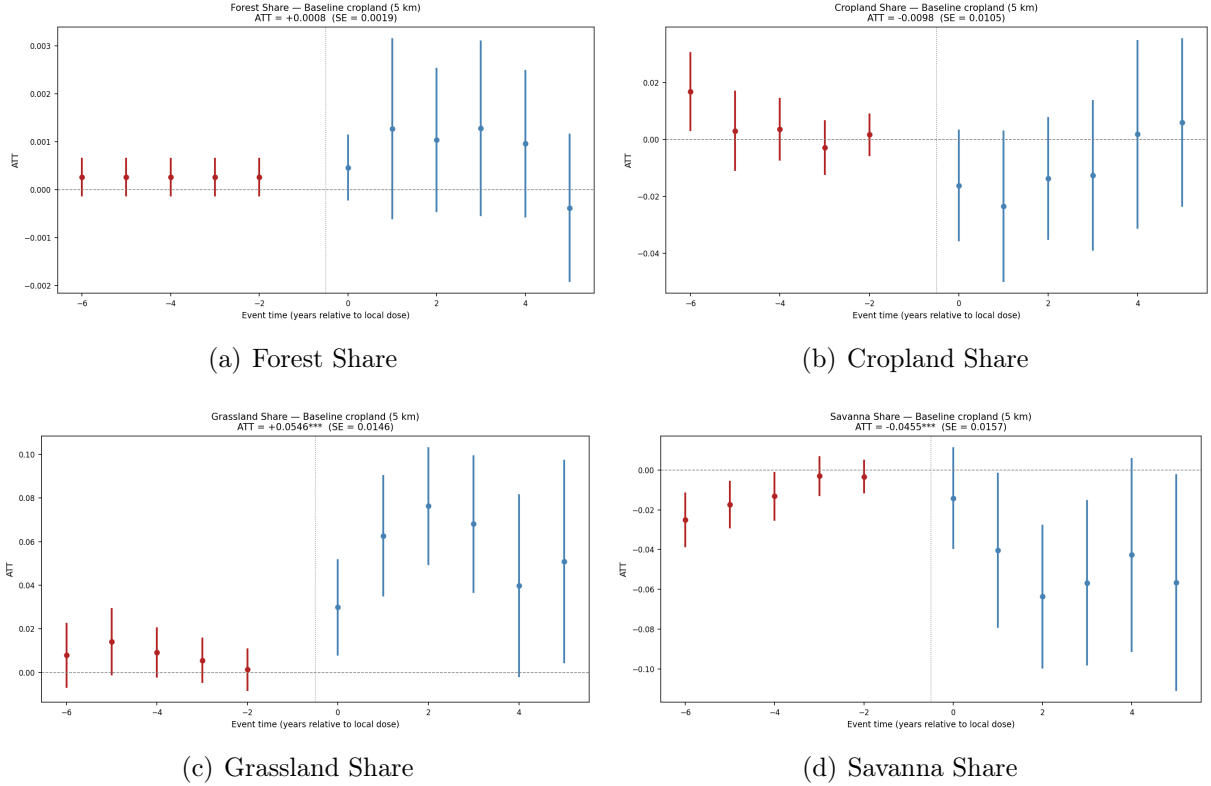


Figure 10: Pooled dose estimator at the pixel level, all Gambela: land-cover class shares on baseline-cropped cells. Design as in Figure 4.

Table 23: Baseline Balance: Field-Verified vs Unverified GIZ Parcels

	Verified	Unverified	Norm. diff	<i>p</i>
Cropland NDVI	0.8226	0.8148	+0.253	0.315
NDVI (raw)	0.8374	0.8289	+0.338	0.008
NPP	0.3607	0.3990	-0.429	0.000
Tree Cover (VCF)	17.9563	17.0399	+0.190	0.131
Forest Share	0.0509	0.0563	-0.033	0.798
Cropland Share	0.0483	0.0819	-0.229	0.058
Grassland Share	0.0256	0.0418	-0.171	0.160
Savanna Share	0.8753	0.8200	+0.233	0.068
Parcel area (ha)	718.9	598.6	+0.326	0.021

Notes: Baseline (2001–2003) pixel means aggregated to the GIZ-parcel level; normalized difference and Welch *t*-test *p*-value, verified ($n = 78$) vs unverified ($n = 252$) parcels (cropland NDVI available for 23/89). Verified parcels are larger, with lower baseline NPP and less baseline cropland — selection toward big, uncultivated savanna tracts, which makes the post-entry cropland conversion on verified parcels conservative. At the household level, exposure to any GIZ parcel is strongly predicted by soil and terrain, but verified- and unverified-exposure profiles are nearly identical; the real-vs-paper contrast is identified conditional on the same observables as the main design.

Table 24: Parallel Trends: Verified vs Unverified GIZ Parcels

Outcome	Placebo gap (joint)		Linear differential	
	Statistic	p	Coefficient	p
NPP	$\chi^2(4) = 2.1$	0.717	+0.00015	0.772
Tree Cover (VCF)	$\chi^2(4) = 5.0$	0.290	−0.05595	0.216
Cropland Share	$\chi^2(4) = 4.3$	0.370	+0.00072	0.784
Forest Share	$\chi^2(4) = 9.4$	0.051	−0.00357***	0.000

Notes: Tests of parallel pre-treatment trends between pixels on field-verified and unverified GIZ parcels. Placebo gap: joint χ^2 on the difference in Callaway–Sant’Anna placebo coefficients at event times -5 to -2 (groups are disjoint pixel sets). Linear test: coefficient on verified $\times$ event-time in pre-treatment observations with pixel and year fixed effects, clustered by parcel. All contrasts are parallel except a linear forest-share differential, which is fully accounted for by prior nearby entries: 75–79% of pixels have an earlier-dated parcel within 5 km before their own parcel’s entry (mean 3.8 years of prior exposure), and on the subsample with no earlier entry within 5 km the differential is zero (+0.0004, $p = 0.73$) — the apparent divergence is deforestation from earlier neighboring doses, not confounding. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

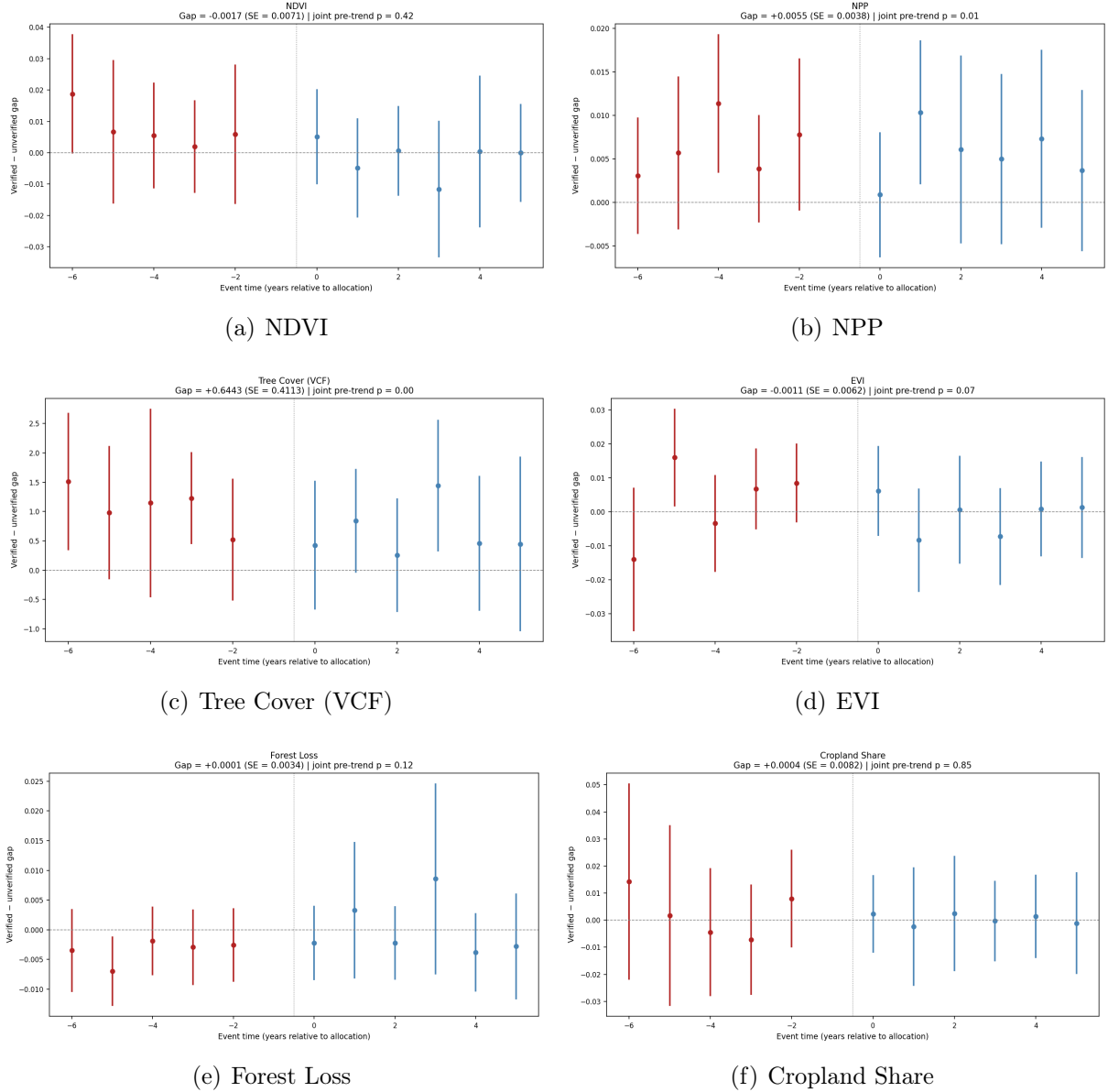


Figure 11: Allocation-aligned event studies for the actual-vs-planned contrast, Gambela. Within each allocation cohort, pixels on field-verified (cultivated) GIZ parcels are compared against pixels on unverified (paper-only) parcels allocated in the same year, with event time relative to the common allocation date; any shock common to an allocation wave differences out. In practice both arms coexist in the dominant 2008 wave (1,913 verified vs. 5,711 unverified pixels). Pre-treatment gaps (red) test that cultivation status is as good as random conditional on allocation timing; joint pre-treatment p -values reported in panel titles alongside overall verified–unverified gaps. Whiskers show 95% CIs; influence-function standard errors clustered by GIZ parcel. Formal balance and trend tests in Tables 23 and 24.

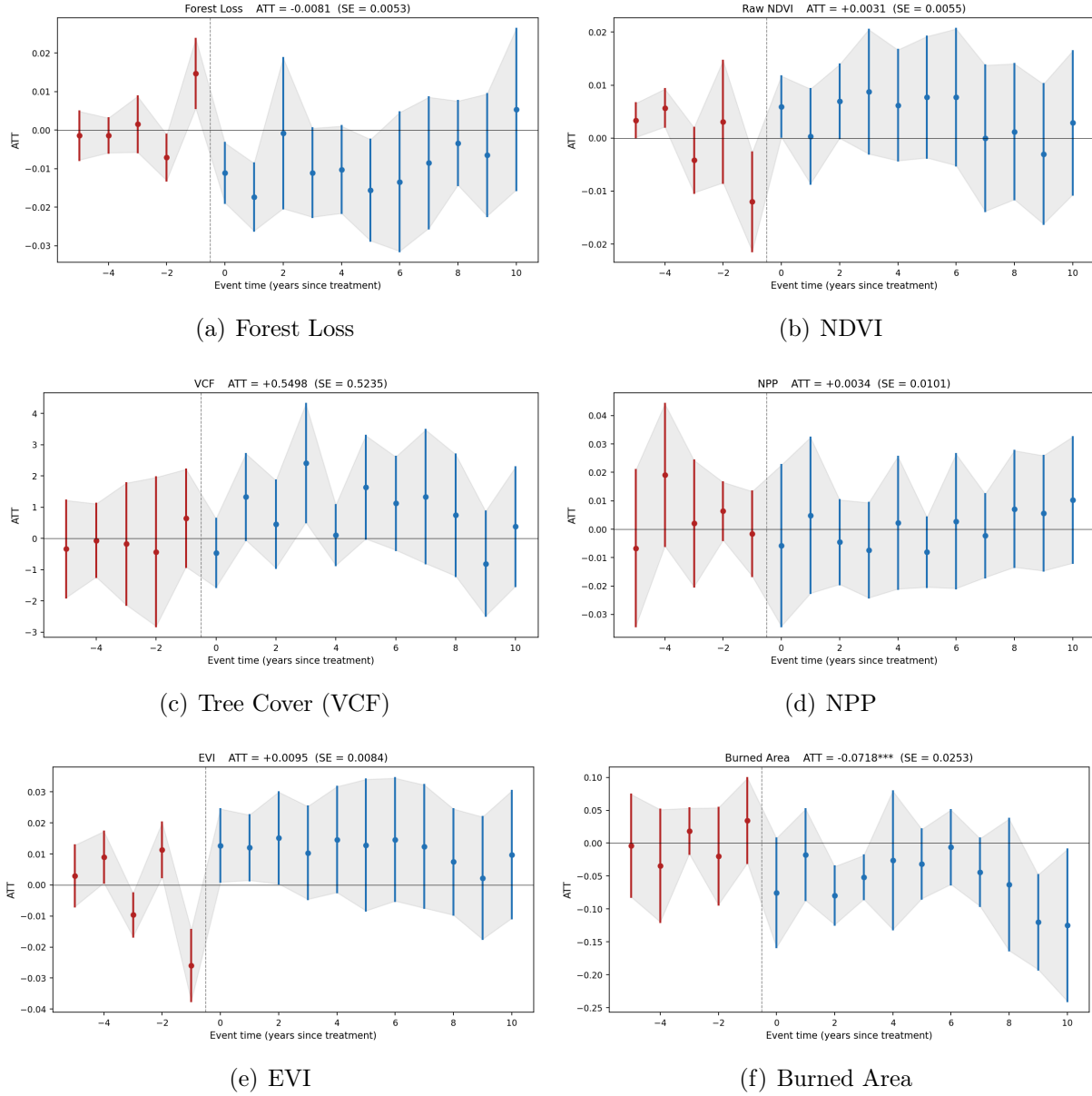
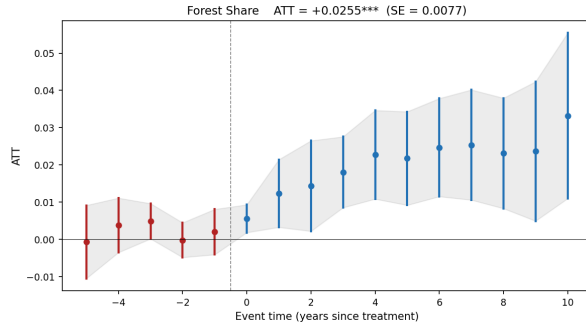
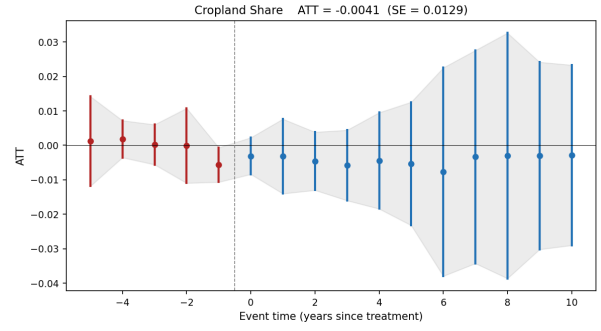


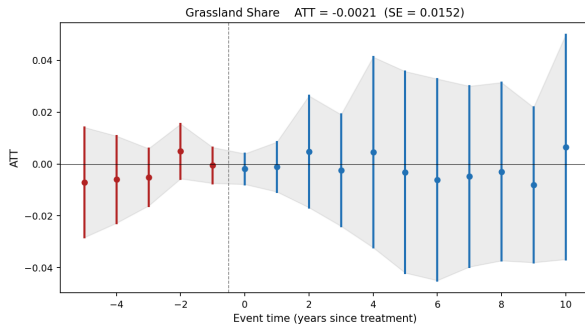
Figure 12: Kebele-level Callaway-Sant'Anna event-study estimates for Gambela region: GEE outcomes. Full sample with never-treated controls; 36 treated kebeles, 162 never-treated. Standard errors clustered at the kebele level.



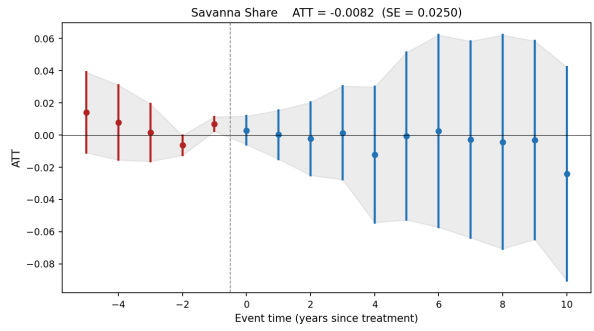
(a) Forest Share



(b) Cropland Share



(c) Grassland Share



(d) Savanna Share

Figure 13: Kebele-level Callaway-Sant'Anna event-study estimates for Gambela region: land cover shares. Full sample with never-treated controls; 36 treated kebeles, 162 never-treated. Standard errors clustered at the kebele level.

Table 25: On-Parcel Effects by GIZ Field-Verified Production Status (Gambella, per additional local dose)

	Verified	Unverified	All
Cropland NDVI	−0.0017 (0.0052)	−0.0037* (0.0021)	−0.0027 (0.0020)
NPP	+0.0011 (0.0014)	−0.0027*** (0.0009)	−0.0056*** (0.0006)
Tree Cover (VCF)	−0.5164* (0.2727)	−0.3208** (0.1283)	−0.6738*** (0.0849)
Burned Area	−0.0312 (0.0211)	−0.1378*** (0.0114)	−0.1322*** (0.0077)
Forest Loss	−0.0132*** (0.0050)	−0.0001 (0.0018)	−0.0002 (0.0013)
Cropland Share	+0.0043 (0.0036)	−0.0049 (0.0045)	−0.0041 (0.0029)
Forest Share	−0.0013 (0.0025)	+0.0046*** (0.0017)	−0.0047*** (0.0010)
Grassland Share	+0.0262*** (0.0070)	+0.0060 (0.0046)	+0.0059* (0.0035)
Switcher pixels ($e=0$)	6762	19068	32655

Notes: Verified: pixel lies in / within 5 km of a GIZ RS4AIM (Eisfelder et al., 2024) LSAI parcel containing at least one 2021–22 ground-truth survey point (96% of on-parcel points record active cropland); Unverified: GIZ parcel never visited by surveyors. Verification is post-treatment and surveyor-selected, so columns are descriptive splits, not causal moderation. Effects are per additional local dose: each k -th dated parcel entry within 5 km is compared against pixels with the identical prior local entry history (entries are spatially clustered and repeated, so own-entry event studies would embed prior nearby doses in their pre-periods). Standard errors are design-based (stratified Neyman, parcels as randomization units within history cells; the sample is the population of on-parcel pixels). Sampling-based parcel-clustered SEs are reported in the replication output. The verified-parcel cropland effect is robust across history-aware designs (+0.025 at first local exposure, +0.014 per local dose; both placebo-clean).

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

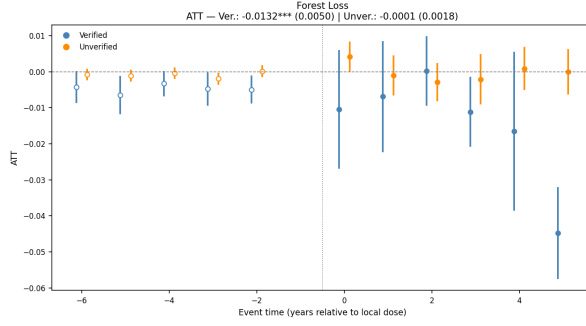
7.5 First-Treatment Event Studies

7.5.1 Gambela Pixel Level

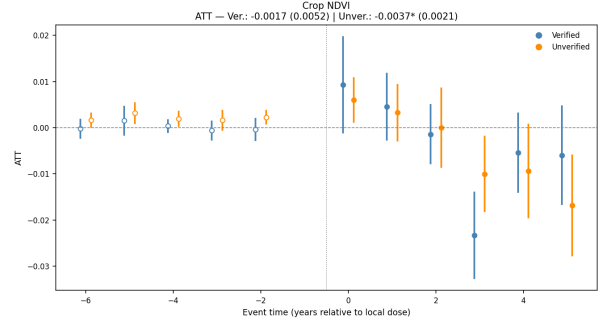
First-treatment Callaway–Sant’Anna pixel event studies (treatment = first LSLT entry within 5 km); the pooled dose estimator in Table 4 is the main specification.

First-treatment Callaway–Sant’Anna event studies; the pooled-dose estimates in the main text nest these exactly at $k = 1$.

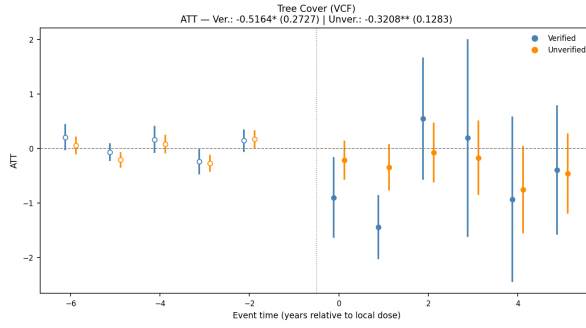
7.6 BJS Imputation Event-Study Estimates



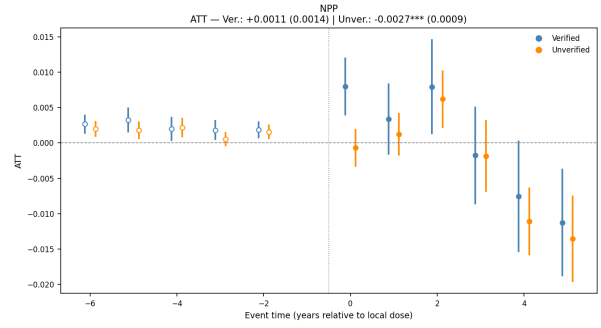
(a) Forest Loss



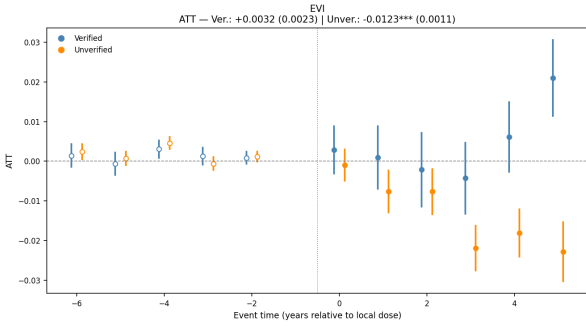
(b) Crop NDVI



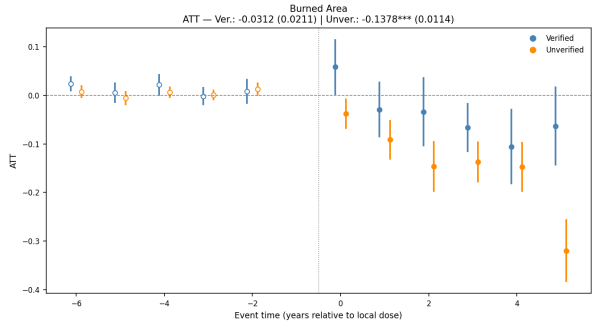
(c) Tree Cover (VCF)



(d) NPP

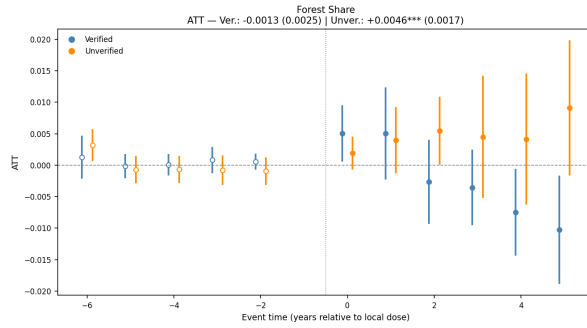


(e) EVI

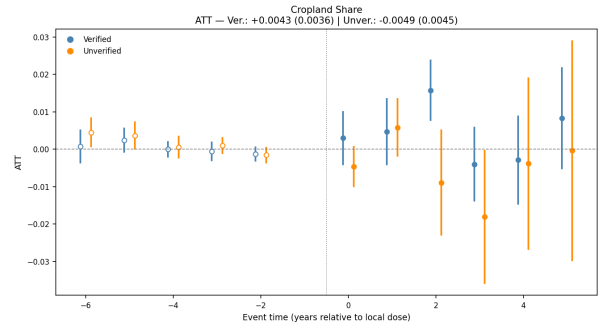


(f) Burned Area

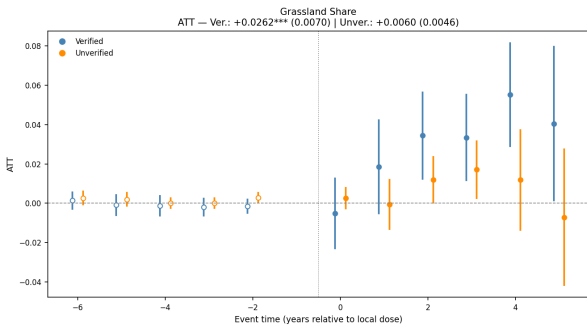
Figure 14: On-parcel event-study estimates by GIZ field-verified production status: GEE outcomes. Verified (blue) and unverified (orange) parcels plotted on common axes; open markers denote pre-treatment periods. Effects are per additional local dose, comparing each k -th local entry (dated parcels within 5 km define the dose sequence) against pixels with the identical prior local history. Whiskers show 95% CIs; overall ATTs and SEs reported in panel titles. Standard errors clustered by parcel.



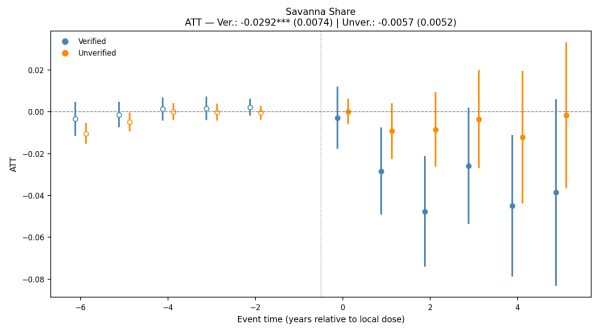
(a) Forest Share



(b) Cropland Share

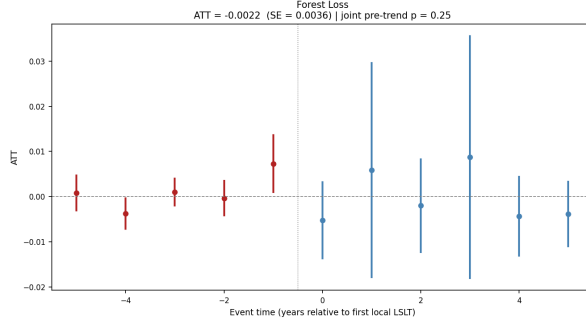


(c) Grassland Share

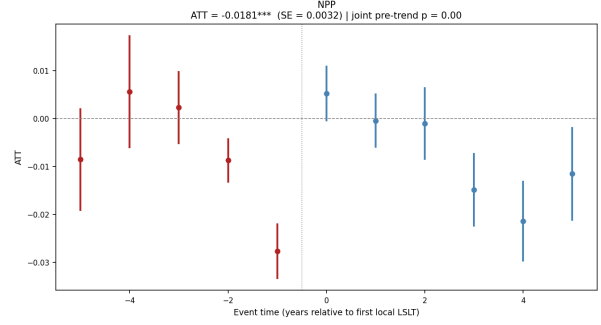


(d) Savanna Share

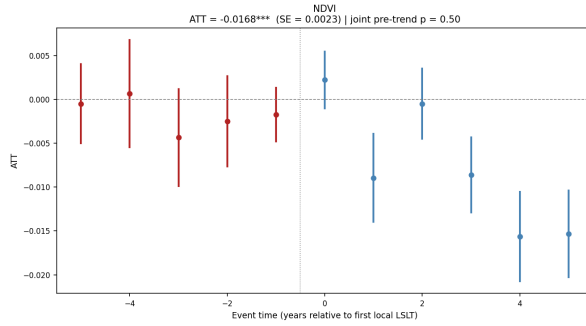
Figure 15: On-parcel event-study estimates by GIZ field-verified production status: land cover shares. Design as in Figure 14.



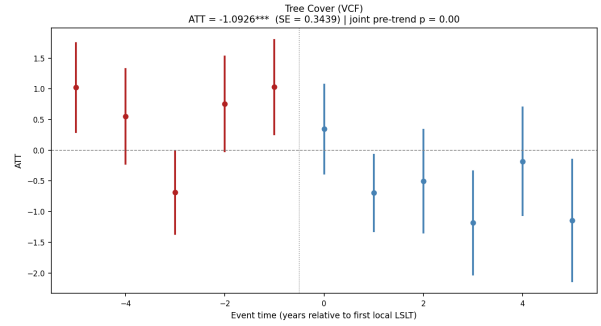
(a) Forest Loss



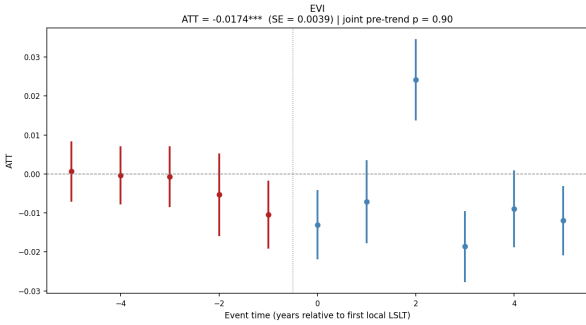
(b) NPP



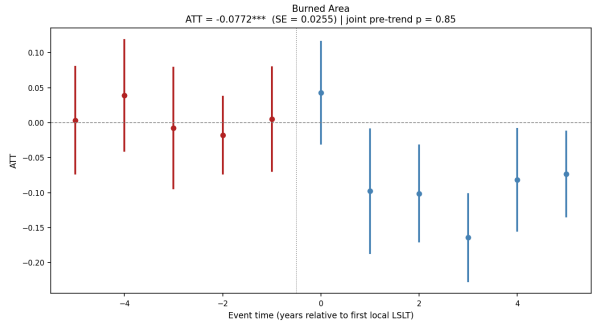
(c) Crop NDVI



(d) Tree Cover (VCF)

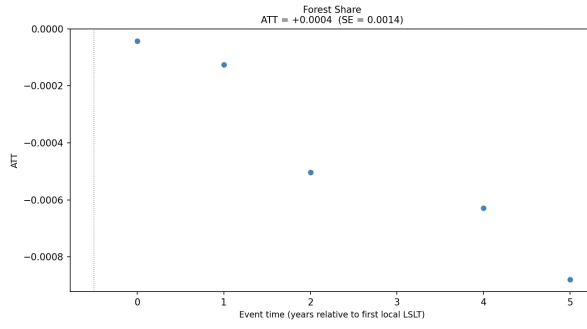


(e) EVI

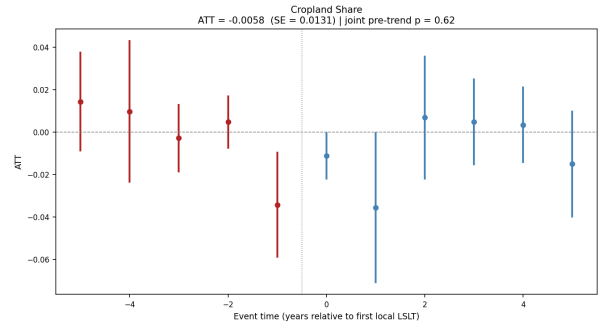


(f) Burned Area

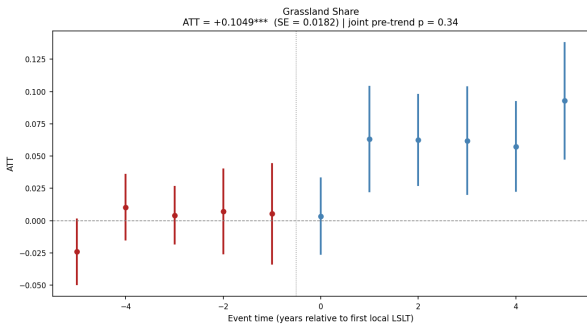
Figure 16: Pixel-level Callaway-Sant'Anna event-study estimates for GEE remote-sensing outcomes. Treatment defined as LSLT entry within 5 km of pixel centroid, restricted to pixels with cropland at baseline (2001); 13,368 cells, parcel-clustered SEs. Whiskers show 95% CIs; overall ATT, SE, and joint pre-treatment placebo p -value reported in each panel title. NPP and Tree Cover fail their placebo tests and should be read with caution.



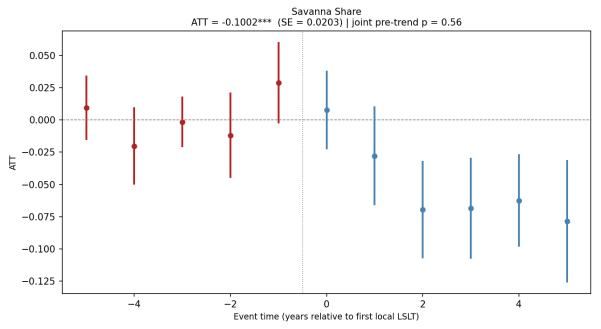
(a) Forest Share



(b) Cropland Share

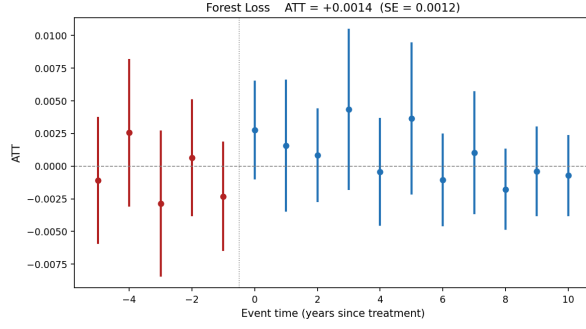


(c) Grassland Share

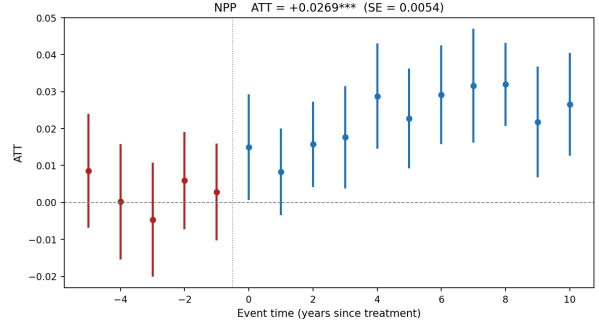


(d) Savanna Share

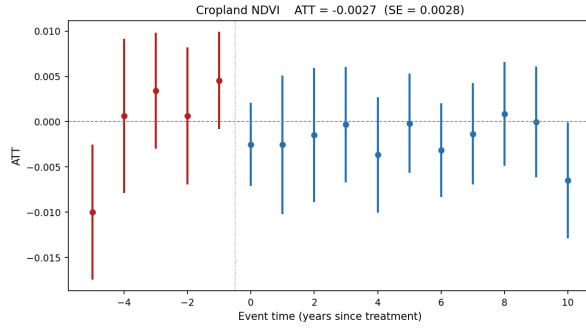
Figure 17: Pixel-level Callaway-Sant’Anna event-study estimates for GEE land classification outcomes. Treatment defined as LSLT entry within 5 km of pixel centroid, restricted to pixels with cropland at baseline (2001); 13,368 cells, parcel-clustered SEs. Whiskers show 95% CIs; overall ATT, SE, and joint pre-treatment placebo p -value reported in each panel title.



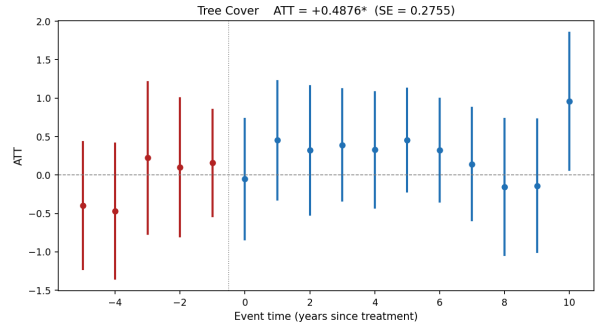
(a) Forest Loss



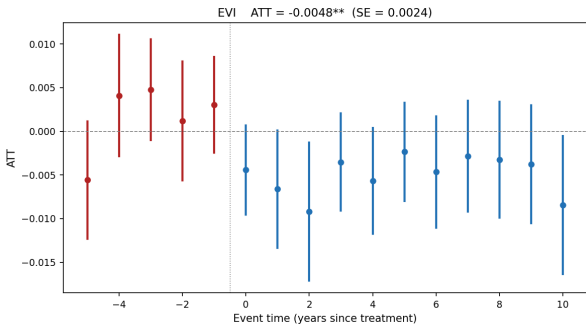
(b) NPP



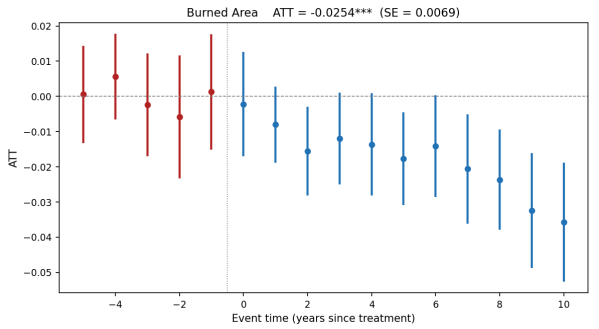
(c) Crop NDVI



(d) Tree Cover (VCF)

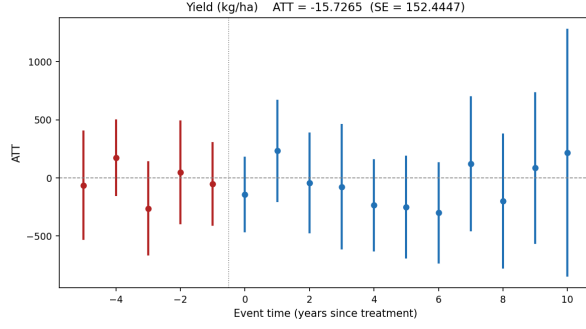


(e) EVI

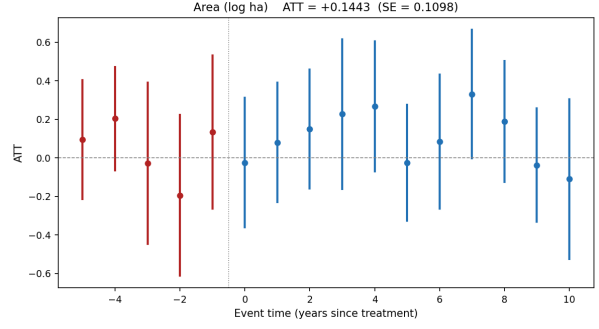


(f) Burned Area

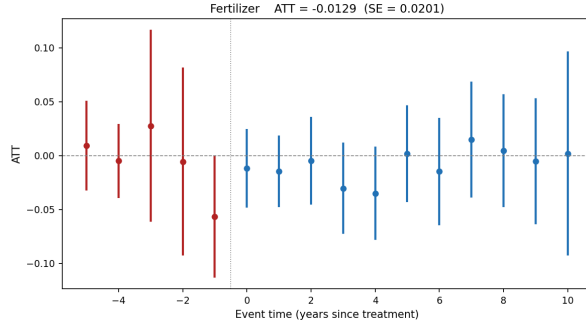
Figure 18: Callaway-Sant'Anna event-study estimates for GEE remote-sensing outcomes. Full sample including never-treated wordas as controls, varying base period. Whiskers show 95% CIs; overall ATT and SE reported in each panel title.



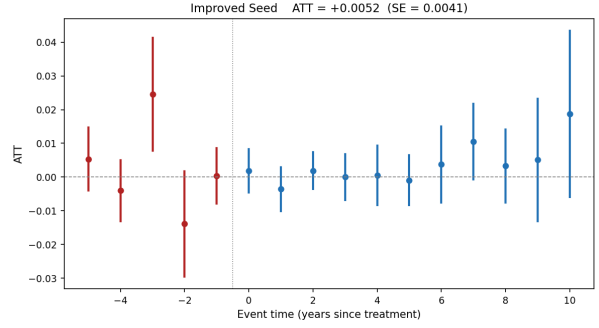
(a) Yield (kg/ha)



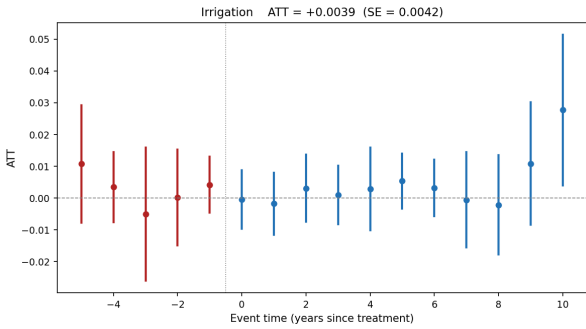
(b) Area (log ha)



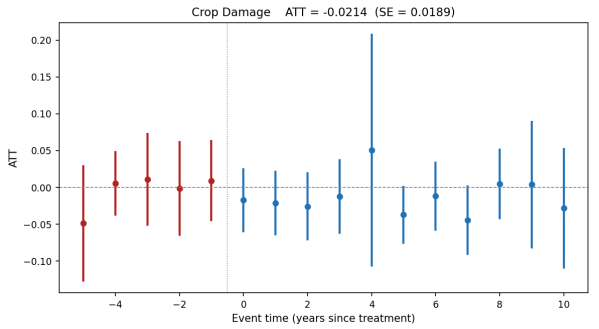
(c) Fertilizer



(d) Improved Seed

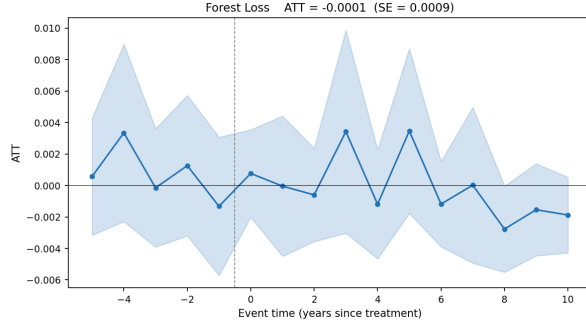


(e) Irrigation

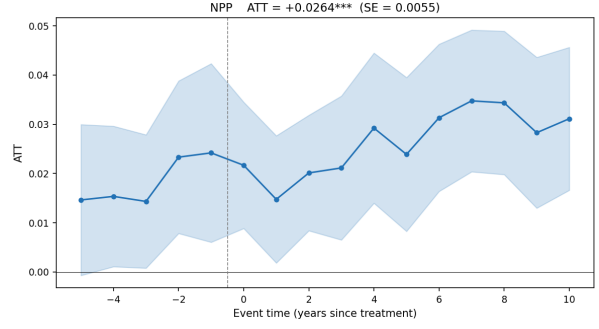


(f) Crop Damage

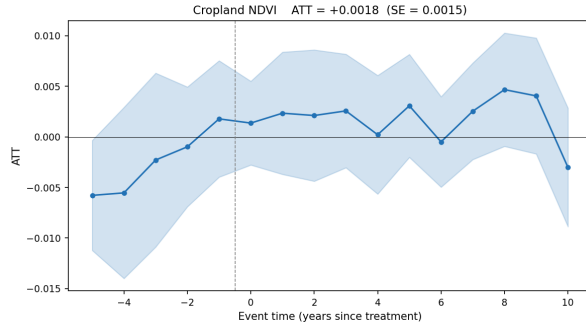
Figure 19: Callaway-Sant'Anna event-study estimates for AgSS agricultural outcomes. Full sample including never-treated woredas as controls, varying base period, repeated cross-section. Whiskers show 95% CIs; overall ATT and SE reported in each panel title.



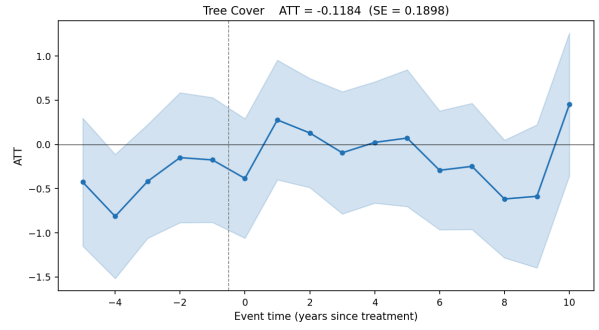
(a) Forest Loss



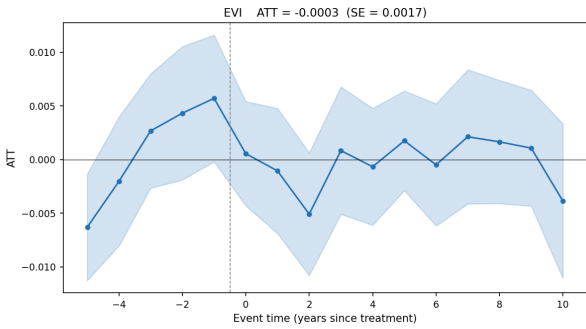
(b) NPP



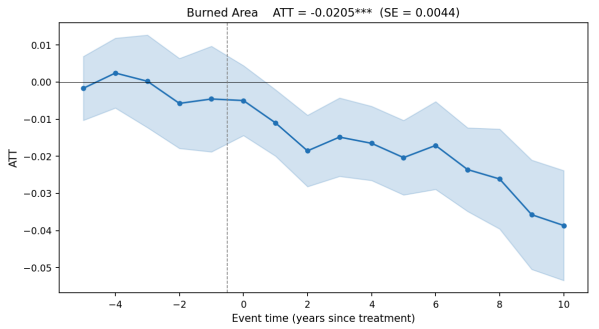
(c) Crop NDVI



(d) Tree Cover (VCF)

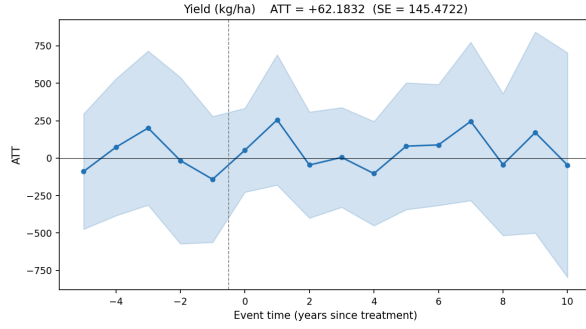


(e) EVI

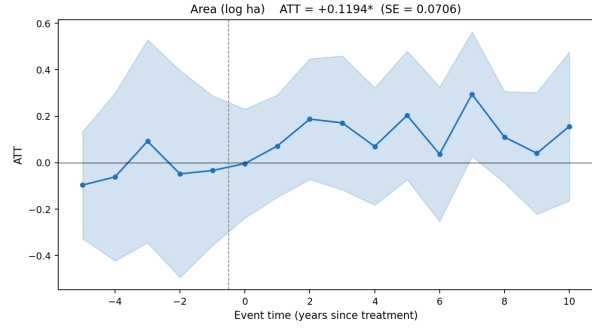


(f) Burned Area

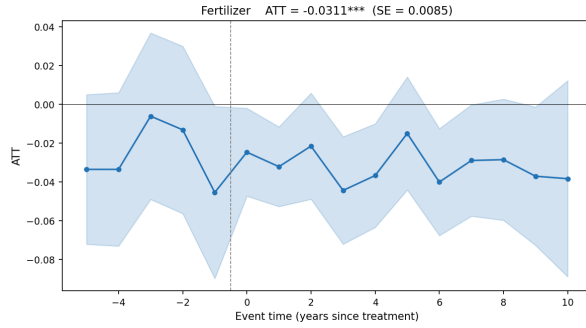
Figure 20: BJS imputation event-study estimates for GEE remote-sensing outcomes. Full sample including never-treated wordas as controls. Whiskers show 95% CIs; overall ATT and SE reported in each panel title.



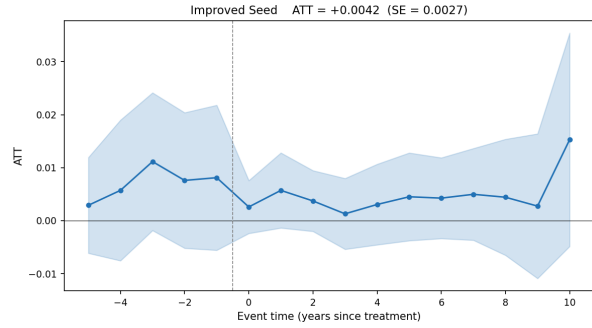
(a) Yield (kg/ha)



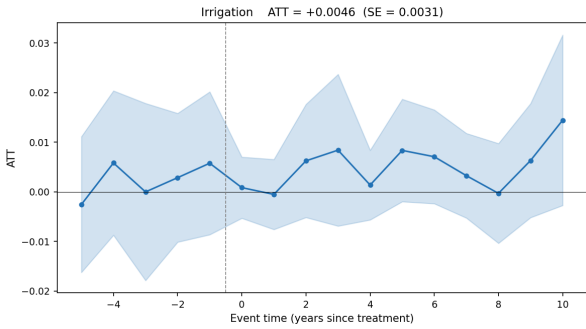
(b) Area (log ha)



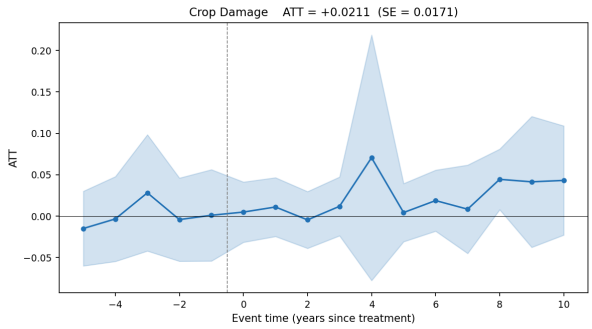
(c) Fertilizer



(d) Improved Seed



(e) Irrigation



(f) Crop Damage

Figure 21: BJS imputation event-study estimates for AgSS agricultural outcomes. Full sample including never-treated wordas as controls. Whiskers show 95% CIs; overall ATT and SE reported in each panel title.